



Modelling Financial Markets

Chapter 13: Machine Learning in Finance



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Acronyms used in this material (1/3)

- ▣ **ADF**: Augmented Dickey–Fuller (test)
- ▣ **AI**: Artificial Intelligence
- ▣ **AIC**: Akaike Information Criterion
- ▣ **AUC**: Area Under the (ROC) Curve
- ▣ **BET**: Bucharest Exchange Trading (index)
- ▣ **BH**: Benjamini–Hochberg (procedure)
- ▣ **CAPM**: Capital Asset Pricing Model
- ▣ **CI**: Confidence Interval
- ▣ **CMA**: Conservative Minus Aggressive (investment factor)
- ▣ **CPCV**: Combinatorial Purged Cross-Validation
- ▣ **CRSP**: Center for Research in Security Prices
- ▣ **CSCV**: Combinatorially Symmetric Cross-Validation
- ▣ **CV**: Cross-Validation
- ▣ **CW**: Clark–West (test for nested models)
- ▣ **DM**: Diebold–Mariano (test)
- ▣ **DML**: Double/Debiased Machine Learning
- ▣ **DSR**: Deflated Sharpe Ratio
- ▣ **ELW**: Exact Local Whittle (estimator)
- ▣ **ES**: Expected Shortfall
- ▣ **EU**: European Union
- ▣ **EV**: Explained Variation
- ▣ **EWMA**: Exponentially Weighted Moving Average
- ▣ **FDR**: False Discovery Rate
- ▣ **FF3**: Fama–French three-factor model
- ▣ **FF5**: Fama–French five-factor model
- ▣ **FFD**: Fixed-width window Fractional Differentiation

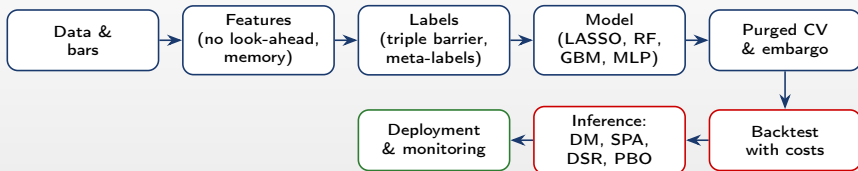
Acronyms used in this material (2/3)

- ▣ **FFN**: FeedForward Network
- ▣ **FN**: False Negative
- ▣ **FRED-MD**: Federal Reserve Economic Data, Monthly Database (McCracken–Ng)
- ▣ **FWER**: Family-Wise Error Rate
- ▣ **GAN**: Generative Adversarial Network
- ▣ **GBM**: Gradient Boosting Machine
- ▣ **GMM**: Generalized Method of Moments
- ▣ **GRS**: Gibbons–Ross–Shanken (test)
- ▣ **GRU**: Gated Recurrent Unit
- ▣ **GW**: Giacomini–White (test)
- ▣ **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ **HML**: High Minus Low (value factor)
- ▣ **IBM**: International Business Machines (Corporation)
- ▣ **IS**: In-Sample
- ▣ **LASSO**: Least Absolute Shrinkage and Selection Operator
- ▣ **LIME**: Local Interpretable Model-agnostic Explanations
- ▣ **LLM**: Large Language Model
- ▣ **LS**: Linear SDF estimated by unregularised Least Squares
- ▣ **LSTM**: Long Short-Term Memory (network)
- ▣ **LW**: Local Whittle (estimator)
- ▣ **MA**: Moving Average
- ▣ **MA20**: 20-day Moving Average
- ▣ **MDA**: Mean Decrease Accuracy
- ▣ **MDI**: Mean Decrease Impurity
- ▣ **ML**: Machine Learning

Acronyms used in this material (3/3)

- ▣ **MLP**: MultiLayer Perceptron
- ▣ **MSCA**: Marie Skłodowska-Curie Actions
- ▣ **OLS**: Ordinary Least Squares
- ▣ **OOS**: Out-Of-Sample
- ▣ **PAC**: Probably Approximately Correct (learning)
- ▣ **PBO**: Probability of Backtest Overfitting
- ▣ **PCR**: Principal Component Regression
- ▣ **PLS**: Partial Least Squares
- ▣ **PSR**: Probabilistic Sharpe Ratio
- ▣ **PT**: Pesaran–Timmermann (test of directional accuracy)
- ▣ **RC**: Reality Check (White's test for data snooping)
- ▣ **RF**: Random Forest
- ▣ **RL**: Reinforcement Learning
- ▣ **RMW**: Robust Minus Weak (profitability factor)
- ▣ **ROC**: Receiver Operating Characteristic (curve)
- ▣ **RSI**: Relative Strength Index
- ▣ **SD**: Standard Deviation
- ▣ **SDF**: Stochastic Discount Factor
- ▣ **SE**: Standard Error
- ▣ **SHAP**: SHapley Additive exPlanations
- ▣ **SPA**: Superior Predictive Ability (test)
- ▣ **SR**: Sharpe Ratio
- ▣ **TN**: True Negative
- ▣ **TP**: True Positive
- ▣ **US**: United States
- ▣ **VIX**: CBOE Volatility Index
- ▣ **WF**: Walk-Forward (validation)

Chapter Route: the Financial ML Workflow



- ▣ Part I: the prediction problem, a minimal toolbox, features and fractional differencing, labels, honest validation
- ▣ Part II: an S&P 500 case study with formal tests, selection and backtest overfitting, machine learning (ML) for inference, a research case study, AI for scientific discovery; technical detail sits in the **Appendix** at the end
- ▣ After this chapter you can: build features and labels without look-ahead; validate with purging and embargo; test a classifier and a backtest against a benchmark (PT, DM, DSR, SPA, PBO)
- ▣ Two process failures [López de Prado \(2018\)](#): most failures come from the process, not the algorithm
 - ▶ **look-ahead bias**: using information not available at decision time
 - ▶ **selection bias**: reporting the best of many trials

Why Machine Learning in Finance, and Why It Is Hard

The promise

- Flexible, non-linear functional forms; interactions found automatically
- Hundreds of candidate predictors (the “factor zoo”, [Chapter 3](#))
- Alternative data: text, images, order books, on-chain data

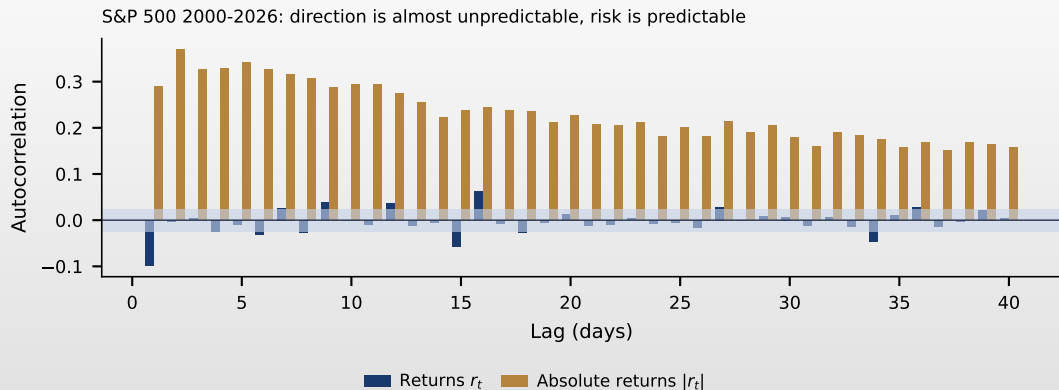
The constraints

- Very low **signal-to-noise ratio**; one serially dependent history that cannot be re-run
- **Non-stationarity**: regimes, breaks; markets react to their users
- Labels must be constructed; errors cost real money
- Massive risk of **overfitting** (fitting noise)

Adaptive Markets Hypothesis: Lo (2004)

- Predictability appears, is arbitrated away and re-appears: a model that worked yesterday is a **hypothesis**, not a law (market efficiency: [Chapter 2](#))
- The hard part is not fitting a model but **not fooling yourself**: correct labels, correct validation, honest backtests

What Is Predictable? Direction vs. Risk (1/2)



What Is Predictable? Direction vs. Risk (2/2)

- Autocorrelation at lag k : $\rho_k = \text{Corr}(x_t, x_{t-k})$, the correlation of a series with its own value k days earlier
 - ▶ x_t : the daily log return r_t or its absolute value $|r_t|$; T : number of observations
 - ▶ light-blue band: $\pm 1.96/\sqrt{T}$, the 95% band under no autocorrelation; a bar outside it is significant at 5%
- Autocorrelations of returns r_t are tiny (inside the band); those of $|r_t|$ are large and persistent
- ML has the best chance where there is structure: **volatility, risk, liquidity, cross-sectional ranking**

 [MFM_ch13_model_comparison](#)

Evidence, and the Economic Value of a Small R_{OOS}^2 (1/2)

- ▣ Gu, Kelly & Xiu (2020): monthly returns of individual US stocks
 - ▶ linear models, penalised regressions, principal component regression (PCR), partial least squares (PLS), forests, boosted trees, neural nets
- ▣ Their fit measure, the out-of-sample R^2 : how much of the squared return a forecast removes, relative to forecasting zero

$$R_{OOS}^2 = 1 - \frac{\sum_t (r_t - \hat{r}_t)^2}{\sum_t r_t^2}$$

- ▶ r_t : realised return in month t of the test period; $\hat{r}_t$: forecast made with information up to $t - 1$
- ▶ the denominator is the error of the zero forecast $\hat{r}_t = 0$
- ▶ $R_{OOS}^2 > 0$: the model beats the zero forecast; $R_{OOS}^2 < 0$: it does worse; at most 1 (perfect forecasts)
- ▣ Result: R_{OOS}^2 is small (well below 1% a month); trees and nets win through **non-linear interactions**; momentum, liquidity and volatility dominate
- ▣ The benchmark matters: Welch & Goyal (2008) show that most published predictors lose to the **historical mean**; Gu, Kelly & Xiu (2020) and Chapter 14 use a zero forecast

Evidence, and the Economic Value of a Small R^2_{OOS} (2/2)

- ▣ Campbell & Thompson (2008): what a small R^2 is worth to an investor; returns have a predictable and an unpredictable part

$$r_{t+1} = \mu + x_t + \varepsilon_{t+1}, \quad R^2 = \frac{\text{Var}(x_t)}{\text{Var}(r_{t+1})}$$

- ▶ μ : mean return; x_t : predictable component, known at t , mean zero; ε_{t+1} : unpredictable shock
- ▶ R^2 : share of the return variance explained by the predictor
- ▣ An unconstrained mean-variance investor who knows x_t (no costs, no estimation error) reaches the Sharpe ratio SR^*

$$SR^{*2} = \frac{SR_0^2 + R^2}{1 - R^2}$$

- ▶ Sharpe ratio (SR): mean excess return divided by its standard deviation
- ▶ SR_0 : SR of buy-and-hold, which ignores x_t ; $R^2 = 0$ gives $SR^* = SR_0$
- ▣ S&P 500, monthly log returns 2001–2026: $SR_0 = 0.128$ a month (0.44 annualised); $R^2 = 0.5\%$ a month $\Rightarrow SR^* = 0.51$ (+15%), $R^2 = 1\% \Rightarrow 0.57$ (+27%)
- ▣ For a **single index** the effective sample is far smaller than for thousands of stocks: the S&P 500 case study tests exactly this

Loss, Penalty and Five Model Families (1/3)

- Supervised learning picks, within a model family, the function that fits the training data best after a penalty for complexity

$$\hat{f} = \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n L(y_i, f(x_i)) + \lambda \Omega(f)$$

- Notation

- ▶ (y_i, x_i) , $i = 1, \dots, n$: target and feature vector of training observation i ; $\mathcal{F}$: the model family (e.g. all trees of depth 3)
- ▶ L : loss, the squared error $(y - f)^2$ or, for a probability f of $y = 1$, the log-loss $-[y \log f + (1 - y) \log(1 - f)]$
- ▶ $\Omega(f)$: complexity penalty; $\lambda \geq 0$: its weight; $\lambda = 0$: no penalty, large λ : a simpler model
- ▶ λ is chosen by K -fold cross-validation (CV): split the data into K folds, fit on $K - 1$, score on the fold left out, average the K scores
- ▶ Expected test error = bias² + variance + noise σ^2 (Hastie, Tibshirani & Friedman (2009), Ch. 7)
 - ▶ bias: error of a too-simple family; variance: sensitivity of $\hat{f}$ to the training sample; σ^2 : variance of the unpredictable part of y
 - ▶ in finance σ^2 dominates, so variance control matters more than flexibility

Loss, Penalty and Five Model Families (2/3)

- ▣ Penalised logit / least squares: the penalty is a mix of absolute and squared coefficients

$$\lambda \Omega(\beta) = \lambda \left[\alpha \|\beta\|_1 + \frac{1-\alpha}{2} \|\beta\|_2^2 \right], \quad \|\beta\|_1 = \sum_j |\beta_j|, \quad \|\beta\|_2^2 = \sum_j \beta_j^2$$

- ▶ $\beta = (\beta_1, \dots, \beta_p)$: coefficients of the p features; $\alpha \in [0, 1]$: share of the absolute-value penalty
 - ▶ Ridge, $\alpha = 0$ (Hoerl & Kennard (1970)): shrinks all coefficients towards zero, none exactly to zero
 - ▶ LASSO, $\alpha = 1$ (Tibshirani (1996)): sets some coefficients exactly to zero, i.e. selects features
 - ▶ Elastic Net, $0 < \alpha < 1$ (Zou & Hastie (2005)): in between
- ▣ The linear benchmark is the logit, the probability of “up” as a function of a linear score

$$p = P(y = 1 \mid x) = \frac{1}{1 + e^{-x^\top \beta}}$$

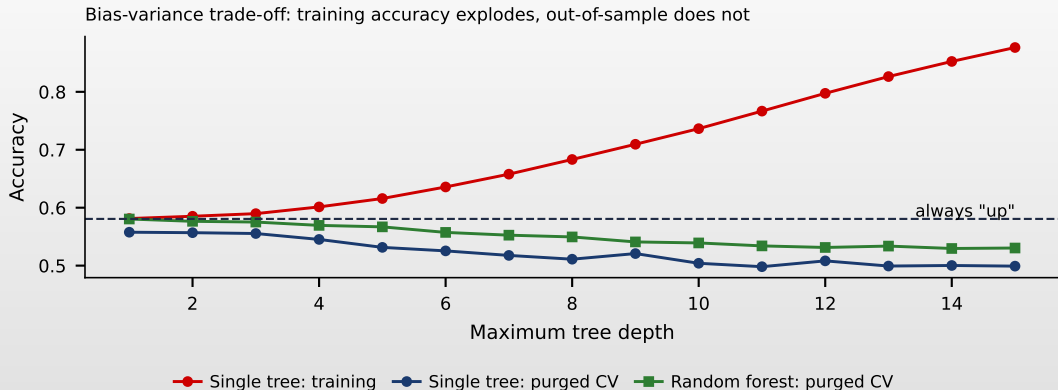
- ▶ $x^\top \beta = \sum_j \beta_j x_j$: linear score; p rises from 0 to 1 as the score rises
- ▶ $\beta_j > 0$: a higher x_j raises the probability of “up”

Loss, Penalty and Five Model Families (3/3)

- Trees: recursive splits that minimise the Gini impurity $G = 1 - \sum_c \hat{\pi}_c^2$ of the resulting nodes
 - ▶ $\hat{\pi}_c$: share of class c in a node; $G = 0$: pure node; $G = 0.5$: two classes 50/50
- Random forest (RF): average of B trees, each grown on a bootstrap sample with a random subset of features at each split
 - ▶ the trees are decorrelated ([Breiman \(2001\)](#))
 - ▶ its out-of-bag error is **optimistic** when labels overlap
- Gradient boosting machine (GBM, [Friedman \(2001\)](#)): $F_m = F_{m-1} + \eta h_m$
 - ▶ F_m : the model after m steps; h_m : a small tree fitted to the negative gradient of the loss (for squared error, the residuals)
 - ▶ $\eta \in (0, 1]$: learning rate; a small η learns slowly and overfits less
- Multilayer perceptron (MLP): $h^{(\ell)} = g(W^{(\ell)}h^{(\ell-1)} + b^{(\ell)})$, $h^{(0)} = x$
 - ▶ $h^{(\ell)}$: output of layer ℓ ; $W^{(\ell)}$, $b^{(\ell)}$: weight matrix and bias vector of layer ℓ
 - ▶ g : ReLU, $g(z) = \max(0, z)$, element by element; early stopping: stop training when the validation loss stops falling; architectures in [Chapter 14](#)
- This chapter classifies the 5-day direction; every CV score is an estimate and is tested against a benchmark



Bias-Variance in Practice: Tree Depth (1/2)

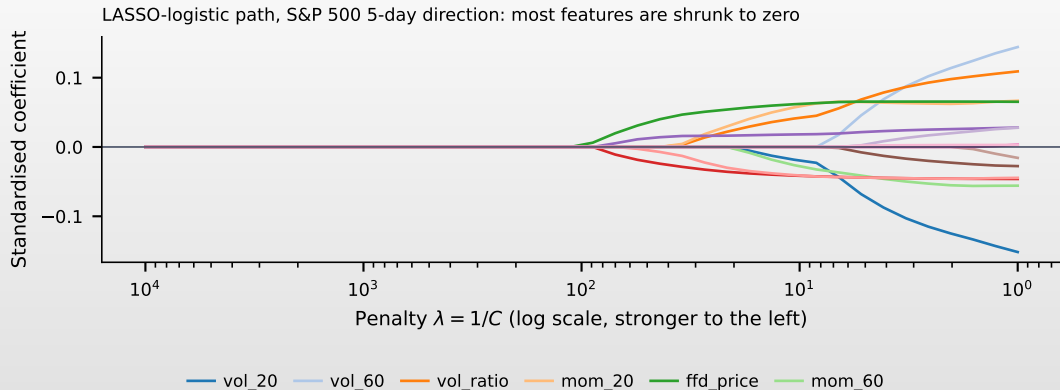


Bias–Variance in Practice: Tree Depth (2/2)

- ▣ S&P 500, 5-day direction; accuracy = share of correctly classified days
- ▣ Training accuracy rises to $\approx 87\%$ with depth; purged-CV accuracy **falls** towards 50%
- ▣ Random forests degrade more slowly, but never beat the “always up” baseline (dashed line)

 MFM_ch13_regularization

LASSO Regularisation Path (1/2)



LASSO Regularisation Path (2/2)

- L_1 -penalised logistic regression for the 5-day direction of the S&P 500, 14 standardised features
- Strong penalty: every coefficient is zero; the first feature to enter is the FFD price, a scaled price level (next section)
- Weak penalty: the volatility features (20- and 60-day volatility and their ratio) get the largest coefficients
- All coefficients are small in absolute value: weak signal, consistent with near-efficient markets

 MFM_ch13_regularization

Stationarity, Memory and the Fractional Difference (1/2)

- ▣ ML inputs need a distribution that does not drift
 - ▶ log prices $X_t = \log P_t$ (P_t : closing price on day t) behave as $I(1)$
 - ▶ returns $\Delta \log P_t = \log P_t - \log P_{t-1}$ are $I(0)$ but **erase the memory** of the price level
- ▣ Integration order: a series X_t is $I(\delta)$ if $(1 - B)^\delta X_t$ is $I(0)$
 - ▶ B : backshift operator, $BX_t = X_{t-1}$, so $(1 - B)X_t = X_t - X_{t-1}$ is the ordinary difference
 - ▶ $I(0)$: stationary with short memory; $I(1)$: a random walk type series, stationary after one difference
 - ▶ δ may be fractional: the series is stationary only for $\delta < \frac{1}{2}$ ([Hosking \(1981\)](#)); $0 < \delta < \frac{1}{2}$: stationary with long memory
- ▣ Idea of fractional differencing: difference by less than 1, removing the unit root while keeping part of the memory

Stationarity, Memory and the Fractional Difference (2/2)

Definition 1 (Fractional difference of order d)

$$\boxed{\tilde{X}_t = (1 - B)^d X_t = \sum_{k=0}^{\infty} w_k X_{t-k}, \text{ with } w_0 = 1 \text{ and } w_k = -w_{k-1} \frac{d-k+1}{k}}$$

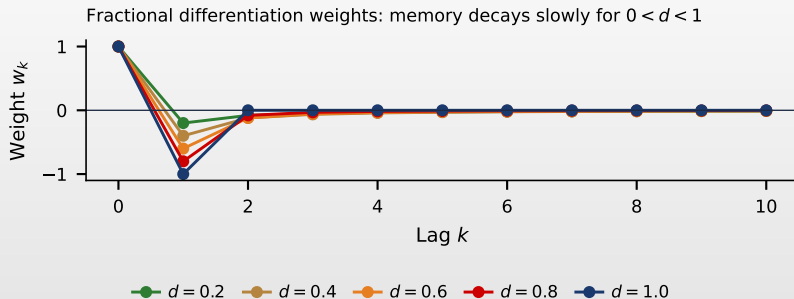
- ▣ d : differencing order chosen by the analyst; k : lag in days; w_k : weight of the value k days back, from the binomial expansion of $(1 - B)^d$
- ▣ $d = 1$: $w = (1, -1, 0, \dots)$, ordinary returns; $0 < d < 1$: weights decay hyperbolically, so memory is kept
- ▣ An $I(1)$ log price differenced with d is $I(1 - d)$: non-stationary for $d \leq \frac{1}{2}$, stationary with long memory for $\frac{1}{2} < d < 1$
- ▣ The infinite weights sum to $(1 - 1)^d = 0$: the level cancels only in the limit (asymptotics in the [Appendix](#))
- ▣ [López de Prado \(2018\)](#) proposes the minimum d at which an ADF test rejects; this section shows why that rule fails

The Fixed-Width Window (FFD)

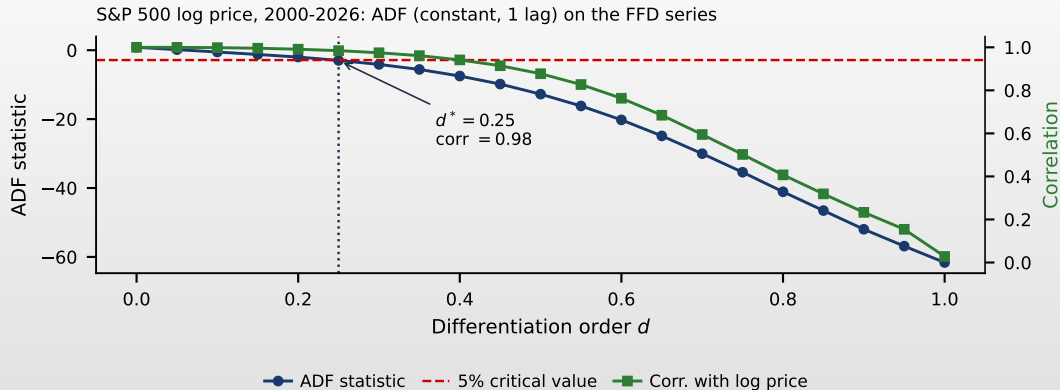
Definition 2 (Fixed-width-window fractional differentiation (FFD))

- ▣ Truncate the weights: keep $w_0, \dots, w_K$ with $|w_k| \geq \tau$ (e.g. $\tau = 10^{-4}$), drop the rest (López de Prado (2018))
- ▣ $\tilde{X}_t = \sum_{k=0}^K w_k X_{t-k}$ for every $t > K$
- ▣ τ : smallest weight kept; K : the last lag kept, i.e. the window length; the first K days have too little history and are dropped
- ▣ Every $\tilde{X}_t$ uses the same number of lags; only past values enter $\Rightarrow$ no look-ahead
- ▣ The smaller d , the more lags are needed: $d = 0.25, \tau = 10^{-4} \Rightarrow K = 444$ lags
- ▣ The truncated weights no longer sum to zero: $\sum_{k=0}^K w_k = 0.178$ at $d = 0.25$ (0.040 at $d = 0.5$)

Fractional Differentiation Weights



- For $d = 1$ only the first lag matters; for small d the weights of distant lags remain non-zero
- All weights after w_0 are negative: $\tilde{X}_t$ compares today's level with a weighted history

The ADF Statistic Along d : S&P 500 (1/2)

The ADF Statistic Along d : S&P 500 (2/2)

- ▣ Augmented Dickey–Fuller (ADF) unit-root test (Time Series course): the t -statistic of γ in
$$\Delta \tilde{X}_t = c + \gamma \tilde{X}_{t-1} + (\text{lagged } \Delta \tilde{X}) + e_t$$
 - ▶ H_0 : $\gamma = 0$, a unit root; more negative values are stronger evidence of stationarity; reject below ≈ -2.86 at 5%
- ▣ FFD log price, S&P 500 2000–2026, constant and 1 lag: the ADF rejects from $d = 0.25$, where $\text{Corr}(\tilde{X}_t, \log P_t) = 0.98$, the correlation of the FFD series with the log price
- ▣ Same series, lag length chosen by AIC (34 lags): $\text{ADF} = -0.72$, no rejection
- ▣ A correlation of 0.98 with an $I(1)$ price is a warning, not a virtue

 MFM_ch13_ffd

Truncated FFD of an $I(1)$ Price Is Still $I(1)$

- Decompose the FFD series:

$$\tilde{X}_t = \sum_{k=0}^K w_k X_{t-k} = \left(\sum_{k=0}^K w_k \right) X_t - \sum_{k=1}^K w_k (X_t - X_{t-k})$$

- ▶ first term: today's level times the sum of the truncated weights; second term: a weighted sum of the k -day changes $X_t - X_{t-k}$
- If X_t is $I(1)$, each $X_t - X_{t-k}$ is $I(0)$; the first term is $0.178 X_t$ at $d = 0.25$: a **scaled random walk**
- The unit-root component has small variance next to the stationary part, so ADF with few lags confuses the two
 - ▶ Monte Carlo: pure random walk, $T = 6,274$ days, FFD with $d = 0.25$: ADF (1 lag) rejects in **97.8%** of 500 samples, at a nominal 5%
- Consequence: the FFD log-price in the case study is a scaled price level, not a stationary feature; the next slides measure the memory directly

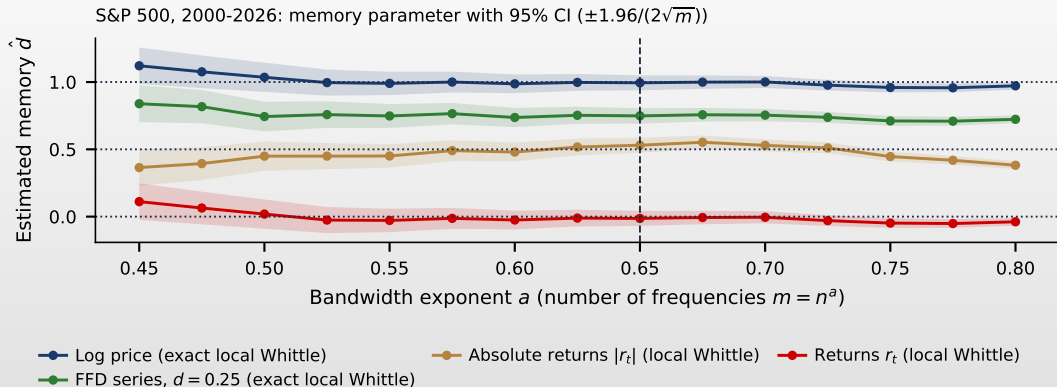
Memory Estimated Directly: Local Whittle (1/3)

- Idea: a long-memory series has most of its variance in slow cycles; the slope of its spectrum near frequency 0 reveals δ

$$f(\lambda) \sim G \lambda^{-2\delta} \quad \text{as } \lambda \rightarrow 0$$

- ▶ $f(\lambda)$: spectral density, the variance contributed by cycles of frequency λ (period $2\pi/\lambda$ days)
- ▶ $G > 0$: scale constant; $\sim$: the ratio of the two sides tends to 1
- ▶ $\delta > 0$: f explodes near 0 (long memory); $\delta = 0$: flat (short memory); $\delta < 0$: f vanishes near 0
- Local Whittle (LW, [Robinson \(1995\)](#)): fits this law only on the first $m = n^a$ Fourier frequencies $\lambda_j = 2\pi j/n$, $j = 1, \dots, m$
 - ▶ n : sample size; $a \in (0, 1)$: bandwidth exponent; a larger a uses more frequencies: less variance, more bias from short-run dynamics
- Exact local Whittle (ELW, [Shimotsu & Phillips \(2005\)](#)): also valid for non-stationary $\delta \geq \frac{1}{2}$; both estimators in the [Appendix](#)
- 95% confidence interval (CI): $\hat{\delta} \pm 1.96/(2\sqrt{m})$; the standard error $1/(2\sqrt{m})$ depends only on m

Memory Estimated Directly: Local Whittle (2/3)

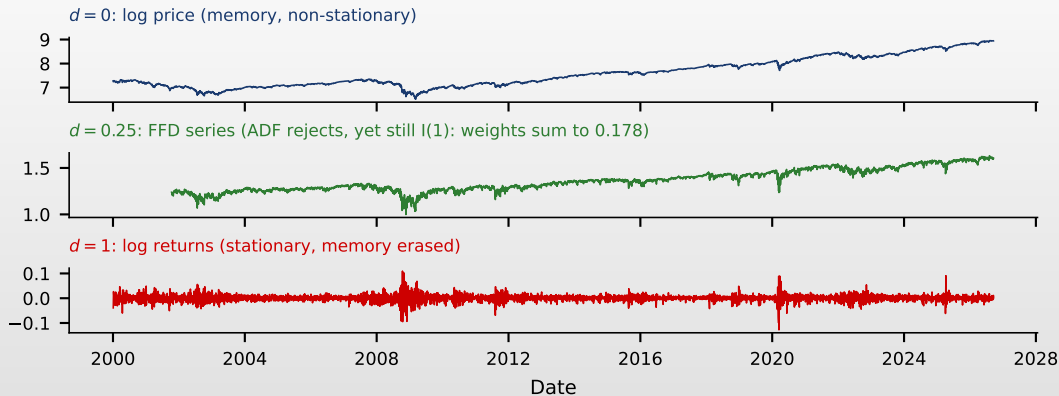


Memory Estimated Directly: Local Whittle (3/3)

- S&P 500, 2000–2026, $m = n^{0.65}$: log price (ELW) 0.99 [0.94; 1.05]; FFD series (ELW) 0.75 [0.69; 0.81]; returns -0.01 $[-0.07; 0.04]$; $|r_t|$ 0.53 [0.47; 0.59]
- The FFD series stays near $1 - d$ over the band (its order is still 1)
 - ▶ the long memory is in **volatility** (Chapters 5 and 9): fractional features belong on $|r_t|$ (Franke, Härdle & Hafner (2019))

 MFM_ch13_ffd

Prices, FFD Series and Returns (1/2)



Prices, FFD Series and Returns (2/2)

- **Top:** log price — $I(1)$
- **Middle:** FFD with $d = 0.25$
 - ▶ the upward trend of the price is still visible: $0.178 \log P_t$ plus a stationary part
- **Bottom:** log returns — only volatility clusters are visible
- The FFD series enters the case study as the feature “FFD log-price”

 MFM_ch13_ffd

From Fixed-Horizon to Triple-Barrier Labels (1/2)

- ▣ **Fixed horizon:** the label is the sign of the h -day log return, with a dead zone around zero

$$r_{t,t+h} = \log P_{t+h} - \log P_t, \quad y_t = \begin{cases} +1 & \text{if } r_{t,t+h} > \tau \\ -1 & \text{if } r_{t,t+h} < -\tau \\ 0 & \text{otherwise} \end{cases}$$

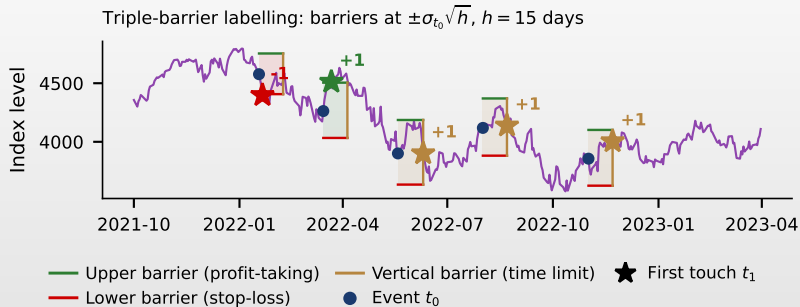
- ▶ P_t : closing price on day t ; h : horizon in days; $\tau \geq 0$: threshold, moves smaller than τ get the label 0
- ▣ Three problems
 - ▶ a constant τ ignores time-varying volatility
 - ▶ the path is ignored: a stop-loss could have been hit before $t + h$
 - ▶ the classes are unbalanced: S&P 500, $h = 10$, $\tau = 0$: 60.5% up vs. 39.5% down

From Fixed-Horizon to Triple-Barrier Labels (2/2)

Definition 3 (Triple-barrier label, López de Prado (2018))

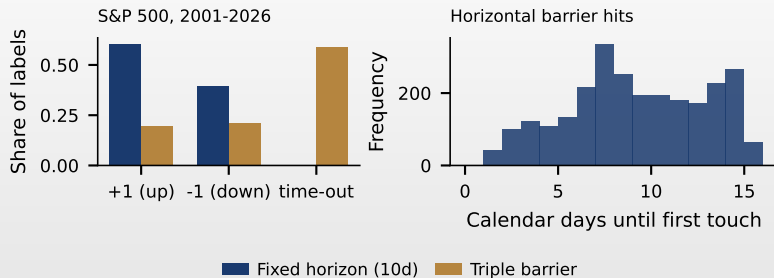
- At event time t_0 , with daily volatility σ_{t_0} and horizon h : upper barrier $\log P_t - \log P_{t_0} \geq \pi_u \sigma_{t_0} \sqrt{h}$ (profit-taking), lower barrier $\leq -\pi_l \sigma_{t_0} \sqrt{h}$ (stop-loss), vertical barrier $t = t_0 + h$
- Label = **first barrier touched** at t_1 : +1 upper, -1 lower; at the vertical barrier $\text{sign}(r_{t_0, t_1})$ (or 0)
- Notation
 - ▶ $\sigma_{t_0} \sqrt{h}$: the typical h -day move; $\pi_u, \pi_l > 0$: barrier widths in units of it
 - ▶ t_1 : day the first barrier is touched; $\text{sign}(x) = +1$ if $x > 0$, -1 if $x < 0$
- σ_{t_0} : square root of the exponentially weighted moving average (EWMA) of squared deviations of daily log returns, span 50
 - ▶ decay factor $\lambda = 1 - 2/51$: each day weighs λ times as much as the following day
 - ▶ the barriers scale with volatility, like a trader's take-profit, stop-loss and maximum holding period
- Each label has a **span** $[t_0, t_1]$: the input of purging in validation

Triple Barrier: S&P 500 in 2022



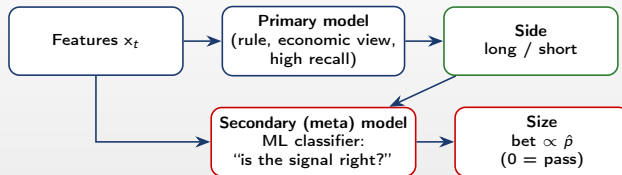
- Five events during the 2022 bear market, $\pi_u = \pi_l = 1$, $h = 15$ days
- January: stop-loss hit (-1); March: profit-taking ($+1$); the others end at the vertical barrier

Label Distribution: Fixed Horizon vs. Triple Barrier



- S&P 500, 2001–2026, $h = 10$: fixed horizon gives 60.5% up / 39.5% down
- Triple barrier ($\pm 1\sigma\sqrt{h}$): 19.6% up, 21.3% down, 59.0% time-out — balanced directional labels
- Right panel: horizontal barriers are typically hit after 1–2 weeks

Meta-Labeling: Deciding Whether to Act (1/2)



- ▣ Meta-label $y_t^{meta} = 1$ if the primary signal was profitable (barrier touched in its direction), 0 otherwise
- ▣ The secondary model estimates $\hat{p} = P(y_t^{meta} = 1 \mid x_t)$, the probability that the signal of day t is right given the features x_t
- ▣ ML learns **when to trust** an interpretable primary rule, not the direction itself

Meta-Labeling: Deciding Whether to Act (2/2)

- Three scores of a filter that decides which signals to act on
 - ▶ precision: share of the trades taken that were profitable
 - ▶ recall: share of the profitable signals that were taken; acting on every signal gives recall 1
 - ▶ $F_1 = 2 \cdot \text{precision} \cdot \text{recall} / (\text{precision} + \text{recall})$: their harmonic mean, between 0 and 1
- Meta-labelling can raise precision at the cost of recall
- Seminar B11, Bitcoin MA20/MA50 crossover, 1,660 test trades: precision $0.475 \rightarrow 0.517$, recall $1 \rightarrow 0.541$, F_1 $0.644 \rightarrow 0.529$; gross return per trade $-0.27\% \rightarrow +0.19\%$
- Filtering need not improve F_1 ; with costs per trade, precision and the return per trade are what matter

Overlapping Labels and Sample Uniqueness (1/2)

- Label i spans $[t_{0,i}, t_{1,i}]$: from its start $t_{0,i}$ to its end $t_{1,i}$; consecutive labels share returns $\Rightarrow$ samples are **not i.i.d.**
- Concurrency: the number of labels whose span contains day t

$$c_t = \sum_i 1\{t_{0,i} \leq t \leq t_{1,i}\}$$

- ▶ $1\{A\}$: indicator, 1 if the condition A holds and 0 otherwise
- Average uniqueness of label i : the average share of each day of its span that belongs to it alone

$$\bar{u}_i = \frac{1}{t_{1,i} - t_{0,i} + 1} \sum_{t=t_{0,i}}^{t_{1,i}} \frac{1}{c_t}$$

- ▶ $\bar{u}_i = 1$: no shared return; $\bar{u}_i = 1/5$: the harmonic mean of c_t over the span of label i is five

Overlapping Labels and Sample Uniqueness (2/2)

Consequences

- Effective sample size $\approx \sum_i \bar{u}_i \ll n$, with n the number of labels (an overlap heuristic, not a variance-based measure)
- Bagging draws near-duplicates; CV scores too optimistic

Remedies

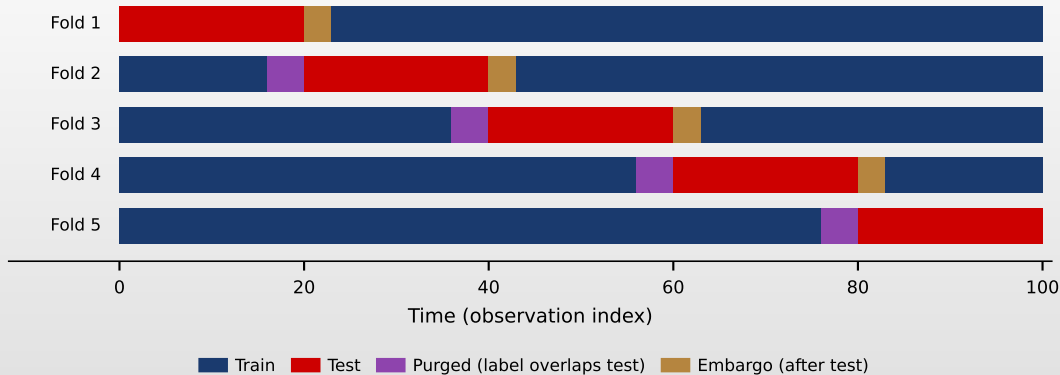
- Weight samples by $\bar{u}_i$ (and by $|r|$)
- Sequential bootstrap; in RF, the fraction of observations drawn for each tree $\approx \bar{u}$, the mean uniqueness

Leakage: What Crosses the Train/Test Boundary

Definition 4 (Leakage)

- The training set contains information that is also present in the test set, so the test score no longer measures performance on genuinely new data
- Serially correlated features ($x_t \approx x_{t+1}$) and overlapping labels ($y_t \approx y_{t+1}$): a shuffled K -fold puts **near-identical** days on both sides
- A flexible model (RF, GBM) memorises the neighbour $\Rightarrow$ excellent “out-of-sample” score; tuning on a leaky CV selects the **most overfitted** model
- Other sources: transformations fitted on the full sample (scaling, principal component analysis, the choice of d for FFD), survivorship bias, data revisions, wrong timestamps

Purged K-Fold with Embargo (1/2)



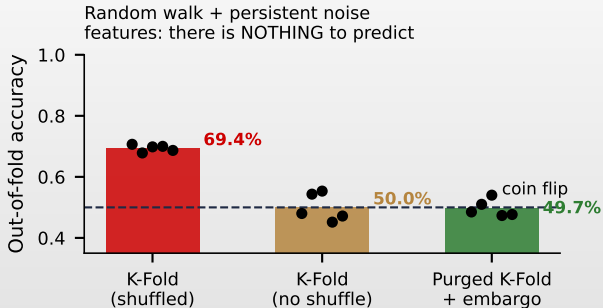
 MFM_ch13_purged_cv

Purged K-Fold with Embargo (2/2)

- Folds are **contiguous** blocks in time (López de Prado (2018)); **purging** removes every training observation whose label span $[t_{0,i}, t_{1,i}]$ overlaps the span of the test labels, from the first test t_0 to the last test t_1
- **Embargo**: also remove a small share of observations right after the test block
 - ▶ e.g. 1%; needed because features look back in time
- One boundary: labels end 2 days after they start, test $\{4, 5\}$: observations 2 and 3 end inside the test block (purged), 6 is the embargo, 7 starts before the last test label ends (purged); full table in the [Appendix](#)

 MFM_ch13_purged_cv

Leakage Experiment: Nothing to Predict



- Price = pure random walk
 - ▶ label = sign of the 20-day forward return
 - ▶ 5 features = 20-day rolling means of independent noise
- True accuracy is 50% by construction; random forest, 5 folds
- **Shuffled K-fold: 69.4%**; contiguous K-fold 50.0%; purged K-fold + embargo 49.7%
- Shuffling fails because labels overlap *and* features persist
 - ▶ a result that looks too good is usually leakage

 MFM_ch13_purged_cv

Transformations on Training Data Only: the FFD Ablation

- The “minimum d ” was chosen on 2000–2026 and then used inside purged CV: the test folds helped choose a feature
 - ▶ the same ADF rule on data up to 2009 gives $d = 0.15$; on the full sample $d = 0.25$
- Case study without the FFD log-price (purged 5-fold, mean over folds)
 - ▶ accuracy: logit 57.5%, RF 56.4%, GBM 55.6%, MLP 55.5%, all below 58.1%
 - ▶ AUC: 0.483, 0.522, 0.514, 0.514
- The negative conclusion does not depend on this feature
- Correct protocol
 - ▶ estimate d (local Whittle, with its CI) inside each training fold only
 - ▶ prefer features with an economic meaning: distances to moving averages, fractional filters of $|r_t|$

Model Selection versus Deployment

- Ordinary K -fold is valid for a purely autoregressive model with serially uncorrelated errors (Bergmeir, Hyndman & Koo (2018)); when it fails: ▶ Appendix
- **Model selection:** purged K -fold, or combinatorial purged CV (CPCV)
 - ▶ CPCV tests on many combinations of groups and returns a **distribution** of Sharpe ratios instead of one number
- **Deployment:** walk-forward (WF)
 - ▶ train on $[0, T_k]$, test on $(T_k, T_{k+1}]$, roll forward (T_k : end of the k -th training window)
 - ▶ realistic, but a single path with high variance
- Rule of thumb: purged K -fold or CPCV to choose, walk-forward for the final, realistic evaluation (details in the ▶ Appendix)

Case Study: Data, Target and Validation

- ▣ S&P 500 index and VIX, daily closes, 2001–2026: $n = 6,269$ observations
- ▣ Target: $y_t = 1\{\log P_{t+5} - \log P_t > 0\}$, equal to 1 if the index is higher 5 days later; label span $[t, t + 5]$ used for purging
- ▣ Validation: purged 5-fold CV, 1% embargo; every model gets the same folds
- ▣ Metrics, against the “always up” baseline
 - ▶ accuracy: share of days whose direction is predicted correctly
 - ▶ receiver operating characteristic (ROC) area under the curve (AUC): the probability that a randomly chosen up day gets a higher predicted probability than a randomly chosen down day
 - ▶ 0.5: no skill, 1: perfect ranking
 - ▶ **fold mean**: the metric computed in each of the 5 test folds, then averaged
 - ▶ **pooled**: the metric computed once on all out-of-fold predictions together; used by the tests
- ▣ Chapter data: daily S&P 500 and VIX closes of the course data

Case Study: 14 Features and Four Models

Features, information up to day t

- **Momentum:** $\log P_t - \log P_{t-h}$, the log return over the last h days, $h \in \{1, 5, 20, 60, 120\}$
- **Realised volatility:** annualised standard deviation (SD) of daily log returns over 20 and 60 days, and their ratio
- **Distance to moving average:** $\log P_t$ minus the mean of $\log P$ over 50 or 200 days
- **RSI(14):** $100 - 100/(1 + RS)$, RS = smoothed average gain over average loss, 14 days; between 0 and 100, high after rises
- **VIX:** log level and 5-day change
- **FFD** log price, $d = 0.25$, chosen on the full sample (ablation in the validation section)

Models, deliberately conservative

- **Logit:** L2-penalised (Ridge), standardised features
- **RF:** 400 trees, minimum leaf 100
- **GBM:** histogram-based, depth 3, learning rate 0.03, 200 trees
- **MLP:** hidden layers of 32 and 16 units, early stopping
- Large leaves and shallow trees limit variance

The Model That Learned “Always Up” (1/2)

	Pred. up	Pred. down
Actual up	TP = 3,572	FN = 68
Actual down	FP = 2,591	TN = 38

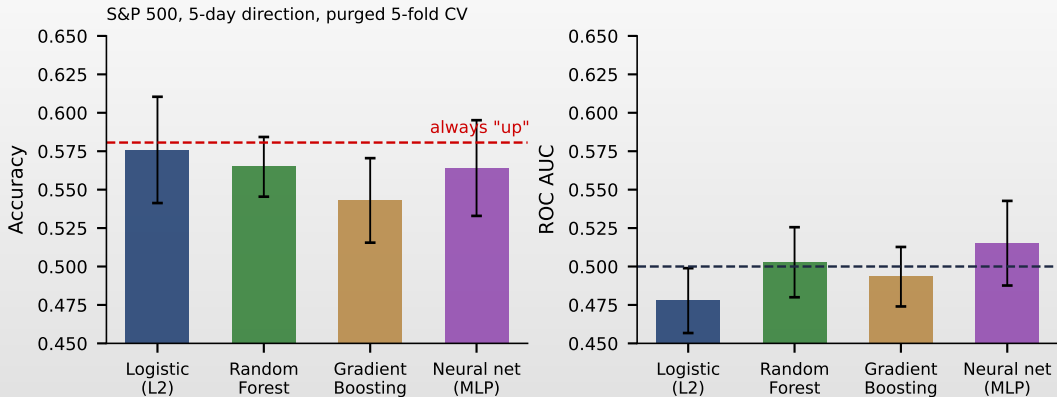
- L2-penalised logit, pooled out-of-fold predictions of purged 5-fold CV, $n = 6,269$
- TP, FN, FP, TN: true positives, false negatives, false positives, true negatives

- Accuracy = $(TP + TN)/n = 57.6\%$: share of days classified correctly
- Precision = $TP/(TP + FP) = 0.580$: share of predicted up days that were up
- Recall = $TP/(TP + FN) = 0.981$: share of actual up days predicted up
- $F_1 = 0.729$: their harmonic mean (formula on the meta-labelling slide)
- The model predicts “up” on 98.3% of days: it has essentially learned “always up”

The Model That Learned “Always Up” (2/2)

- ▣ **Majority baseline** from the same table: actual up days $(3,572 + 68)/6,269 = 3,640/6,269 = 58.1\%$
 - ▶ the accuracy of predicting “up” every day, with no model at all
- ▣ A 57.6%-accurate model is *worse* than doing nothing
- ▣ Other baselines: yesterday’s direction, a moving-average rule, buy-and-hold for trading
- ▣ A model has value only if it beats the baseline **out of sample**, after costs, by more than sampling error

Model Comparison, Fold Means (1/2)



Model Comparison, Fold Means (2/2)

Fold mean	Logit	RF	GBM	MLP
Accuracy	57.6%	56.5%	54.3%	56.4%
AUC	0.478	0.503	0.493	0.515

- Bars: mean over the 5 purged folds; whiskers: ± 1 SD across folds
- Baseline “always up”: **58.1%**; no model beats it

 [MFM_ch13_model_comparison](#)

Are the Differences Significant? Pesaran–Timmermann and AUC Intervals (1/2)

- ▣ **Pesaran–Timmermann** (PT, Pesaran & Timmermann (1992)): are predicted and realised directions related?

$$PT = \frac{\hat{P} - \hat{P}^*}{\widehat{SE}} \xrightarrow{d} N(0, 1), \quad \hat{P}^* = \hat{p}_y \hat{p}_x + (1 - \hat{p}_y)(1 - \hat{p}_x)$$

- ▶ $\hat{P}$: share of days with the correct sign; $\hat{p}_y$, $\hat{p}_x$: shares of realised and of predicted “up”
- ▶ $\hat{P}^*$: share correct expected if predictions were independent of outcomes; $\widehat{SE}$: standard error of $\hat{P} - \hat{P}^*$ under H_0 (▶ Appendix)
- ▶ H_0 : independence; $\xrightarrow{d} N(0, 1)$: approximately standard Normal in large samples; $PT > 1.645$: skill at 5% (one-sided); $PT < 0$: fewer hits than chance
- ▣ With overlapping labels: slope t in $y_t = a + b x_t + u_t$, x_t the predicted and y_t the realised direction, Newey–West errors (Newey & West (1987)); $b > 0$: a predicted “up” is followed by more “up”
- ▣ **AUC**: the DeLong standard error (SE) assumes independent days (▶ Appendix)
 - ▶ a circular block bootstrap resamples blocks of $L = 20$ consecutive days (Künsch (1989)) and keeps the serial dependence

Are the Differences Significant? Pesaran–Timmermann and AUC Intervals (2/2)

Pooled out-of-fold	Logit	RF	GBM	MLP
$\hat{P} / \hat{P}^*$	0.576 / 0.578	0.565 / 0.570	0.543 / 0.557	0.564 / 0.565
$PT / HAC\ t\ (\text{lag } 4)$	-1.28 / -1.00	-1.61 / -1.16	-3.07 / -2.31	-0.36 / -0.26
AUC	0.479	0.495	0.490	0.512
block-bootstrap 95% CI	[0.455; 0.504]	[0.472; 0.520]	[0.468; 0.514]	[0.489; 0.537]

- HAC t : the slope t -statistic with heteroskedasticity and autocorrelation consistent (HAC) errors, 4 lags
- No significant positive directional skill; every AUC interval contains 0.5; GBM is significantly worse than independence
- Consistent with weak-form efficiency ([Chapter 2](#)); look at cross-sections ([Gu, Kelly & Xiu \(2020\)](#)), risk targets, crypto delisting ([Będowska-Sójka, Wójcik & Pele \(2026\)](#)), text ([Chapter 15](#))

Comparing Forecasts: Diebold–Mariano, Clark–West, Giacomini–White (1/2)

- ▣ Two probability forecasts of $y_t \in \{0, 1\}$: model 1 gives $\hat{p}_{1,t}$, the benchmark (model 0, e.g. a constant) gives $\hat{p}_{0,t}$
- ▣ Loss of a forecast $\hat{p}_t$, lower is better: the log-loss of the toolbox slide, with $f = \hat{p}_t$, or the Brier loss $(y_t - \hat{p}_t)^2$
- ▣ Loss differential $d_t = L(y_t, \hat{p}_{1,t}) - L(y_t, \hat{p}_{0,t})$; $d_t > 0$: model 1 worse on day t
- ▣ **Diebold–Mariano** (DM, [Diebold & Mariano \(1995\)](#)): is the mean loss differential zero?

$$DM = \frac{\bar{d}}{\sqrt{\hat{\Omega}_{NW}/n}} \xrightarrow{d} N(0, 1) \quad \text{under } H_0 : \mathbb{E}[d_t] = 0$$

- ▶ $\bar{d}$: mean of d_t over the n test days; $\hat{\Omega}_{NW}$: HAC long-run variance of d_t , by Newey–West
- ▶ $DM > 1.96$: model 1 significantly worse at 5%; $DM < -1.96$: significantly better

Comparing Forecasts: Diebold–Mariano, Clark–West, Giacomini–White (2/2)

- ▣ **Nested models** (the logit with 14 features nests the constant, $\beta = 0$): under H_0 the large model estimates zeros with noise, $\mathbb{E}[d_t] > 0$, and DM rarely favours it

- ▣ **Clark–West** (CW, [Clark & West \(2007\)](#)): one-sided HAC t -test of $\mathbb{E}[f_t] = 0$ against $\mathbb{E}[f_t] > 0$

$$f_t = (y_t - \hat{p}_{0,t})^2 - [(y_t - \hat{p}_{1,t})^2 - (\hat{p}_{0,t} - \hat{p}_{1,t})^2]$$

- ▶ the Brier-loss gain of model 1, plus the term $(\hat{p}_{0,t} - \hat{p}_{1,t})^2$ that removes the noise cost of estimating zero coefficients
 - ▶ a large positive t : the features add information
 - ▶ derived for nested linear models estimated by OLS; for penalised-logit probabilities an approximation, not a theorem
- ▣ **Giacomini–White** (GW, [Giacomini & White \(2006\)](#)): tests the forecasting *method* as deployed
 - ▶ with a bounded estimation window; CW asks whether the population model adds information
 - ▶ statistic in the [Appendix](#)

Feature Importance: MDI, MDA and SHAP

Mean decrease impurity (MDI)

- Average Gini reduction from splits on a feature, over the trees
- In sample, trees only; favours continuous, high-cardinality features

Mean decrease accuracy (MDA)

- Drop in an out-of-sample score (here AUC) when a feature is randomly permuted
- Any model; can be negative: the feature hurts

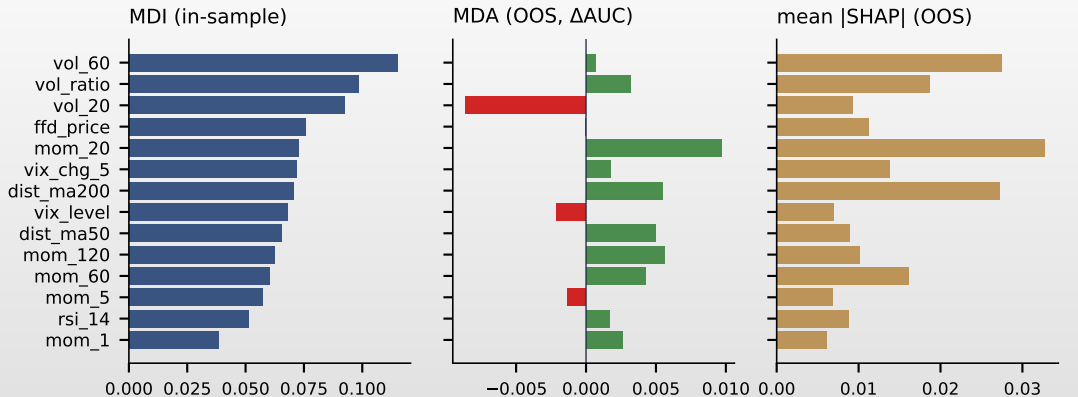
SHAP (Lundberg & Lee (2017))

- Shapley value ϕ_j : average marginal contribution of feature j over coalitions
- Additive: $\hat{f}(x) = \phi_0 + \sum_j \phi_j$, ϕ_0 the average prediction; mean $|\phi_j|$ = global importance

- MDI answers “what did the model use?”; MDA and SHAP answer “what helps on new data?”
- Correlated features share and dilute importance (substitution): cluster them first; coalition formula and the conditional vs. interventional choice in the [Appendix](#)

 Shapley Values

Feature Importance in the Case Study (1/2)



MFM_ch13_feature_importance

Feature Importance in the Case Study (2/2)

- Fit on the first 70%, evaluated on the last 30% after a purge
 - ▶ **RF** gives MDI and MDA, **GBM** gives SHAP: the differences are not purely methodological
- MDI ranks volatility first; MDA and SHAP point to 20-day momentum and the distance from the 200-day moving average
- 20-day volatility: high MDI, **negative** MDA: a sign of overfitting, not a proof
- All importances are tiny ($\Delta\text{AUC} < 0.01$); how to read them: [▶ Appendix](#)
- Correlated features: permutation creates off-manifold points, so MDA and SHAP **extrapolate** ([Hooker, Mentch & Zhou \(2021\)](#))

 MFM_ch13_feature_importance

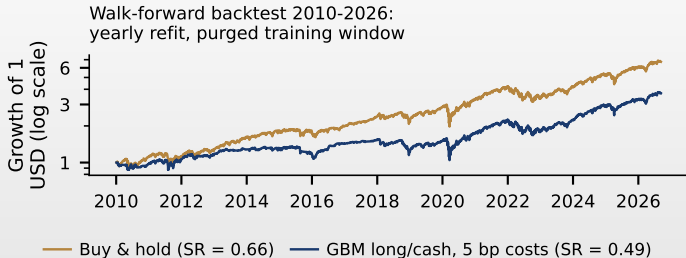
Walk-Forward Backtest: Strategy and Design

- Position $w_t \in \{0, 1\}$: long the S&P 500 if $\hat{p}_t > 0.5$, otherwise cash, decided with information up to t ; $\hat{p}_t$: the model's probability of "up"
- Strategy return: the index return when invested, minus a cost each time the position changes

$$r_{t+1}^S = w_t r_{t+1} - c |w_t - w_{t-1}|$$

- ▶ r_{t+1} : daily log return of the index; $r - c$ is the first-order approximation of $\log(e^r - c)$
- ▶ turnover $|w_t - w_{t-1}|$: 1 on the days the position changes; cost $c = 5$ bp (0.05%) per unit traded
- ▶ exposure $\bar{w} = \frac{1}{T} \sum_t w_t$: share of the T days spent invested
- Model: GBM refitted every January on all past data, purging labels that overlap the test year
- Out-of-sample period 2010–2026, one refit a year; benchmark buy-and-hold ($w_t = 1$: exposure 100%, no turnover)
- The only decision the backtest evaluates is the one a trader could have taken on day t

Walk-Forward Backtest: Results



- 2010–2026: strategy Sharpe **0.49** vs. buy-and-hold **0.66** (annualised); average exposure 77%; the model mostly learned “be long” and exited at the wrong times
- Difference of the two Sharpe ratios -0.17 (Lo SE ≈ 0.25 each, formula in [Sharpe Ratio Inference](#)): Ledoit–Wolf HAC SE 0.095 , $p = 0.073$ ([Ledoit & Wolf \(2008\)](#)); block-bootstrap CI $[-0.36; 0.02]$: no evidence of skill, and no significant evidence that the strategy is worse
- With $d = 0.15$, chosen by the same rule on data up to 2009, SR = 0.43; without the FFD log-price 0.51

Walk-Forward: Do the Forecasts Beat a Constant?

- Walk-forward 2010–2026, $n = 4,198$ days, 60.2% “up”; benchmark: the constant forecast = training share of “up”
- Logit (nests the constant): log-loss 0.6822 vs. 0.6757
 - ▶ DM $t = 2.39$ ($p = 0.017$): the estimated logit is significantly *worse*
 - ▶ Clark–West $t = -0.21$ (one-sided $p = 0.58$): no evidence that the features carry information
 - ▶ Giacomini–White = 5.77 (χ^2_2 , $p = 0.056$): no significant difference at 5%; approximate, since the window expands (rolling-window result in the [Appendix](#))
- GBM: log-loss 0.6971; DM $t = 5.09$, GW = 25.8 ($p < 0.001$): worse than a constant; Pesaran–Timmermann -0.74
- An honest negative result is a result: no model beats “always up” or a constant forecast

Selection Bias and the False Strategy Theorem

- A researcher tries N strategies or configurations and reports the best; even if all have zero true Sharpe ratio, $\max_n \widehat{SR}_n$ grows with N : **selection bias**
- Harvey, Liu & Zhu (2016): given hundreds of published factors, a new factor needs $t > 3$, not 2

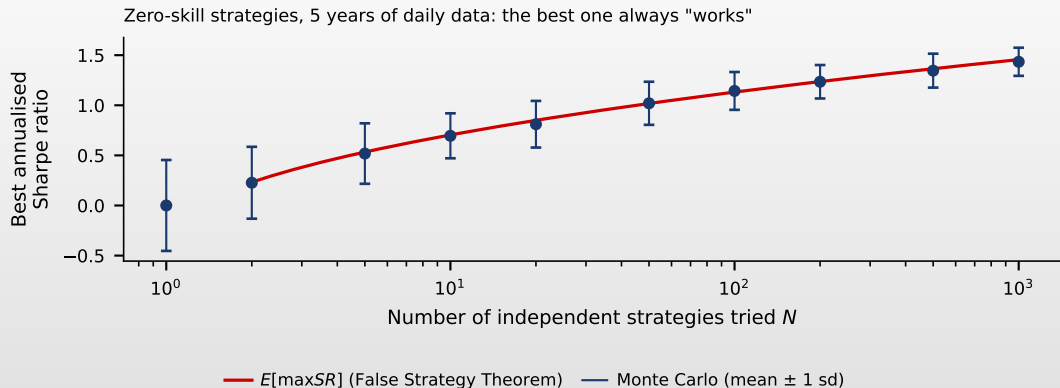
Theorem 1 (False Strategy Theorem, Bailey & López de Prado (2014))

- If $\widehat{SR}_n$, $n = 1, \dots, N$, are independent and approximately $N(0, \text{Var}[\widehat{SR}])$ (no skill), the **expected maximum Sharpe ratio** is

$$SR_0 \equiv \mathbb{E} \left[\max_n \widehat{SR}_n \right] \approx \sqrt{\text{Var}[\widehat{SR}]} \left[(1 - \gamma) \Phi^{-1} \left(1 - \frac{1}{N} \right) + \gamma \Phi^{-1} \left(1 - \frac{1}{Ne} \right) \right]$$

- Notation
 - ▶ $\widehat{SR}_n$: estimated Sharpe ratio of strategy n ; $\text{Var}[\widehat{SR}]$: sampling variance of one estimated SR; i.i.d. returns: $\approx 1/T$ per period, T the number of return periods
 - ▶ Φ : standard Normal distribution function; Φ^{-1} : its quantile function; $e \approx 2.718$; $\gamma \approx 0.5772$: Euler–Mascheroni constant
- Reading: SR_0 is the Sharpe ratio the luckiest of N worthless strategies is expected to show; it rises with N and with the sampling noise $\sqrt{\text{Var}[\widehat{SR}]}$

The Best of N Zero-Skill Strategies (1/2)

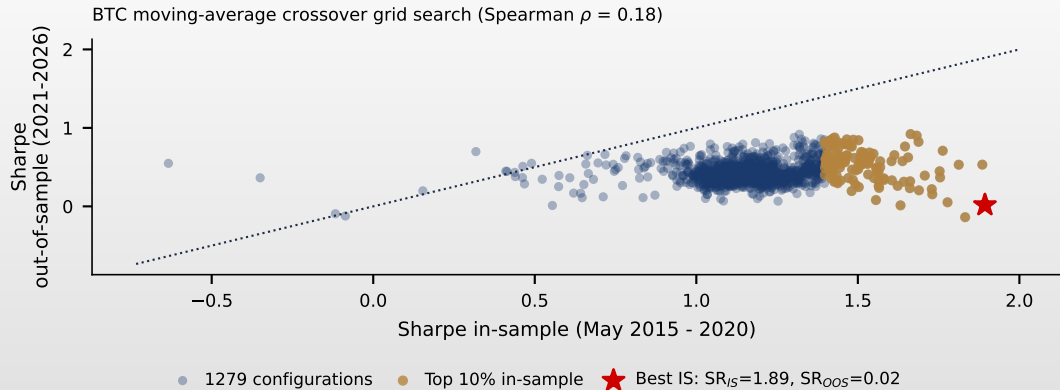


The Best of N Zero-Skill Strategies (2/2)

- ▣ Strategies with i.i.d. zero-mean returns, 5 years of daily data (annualised $\text{Var}[\widehat{SR}] \approx 1/5$)
- ▣ Trying $N = 1000$ variants yields a best Sharpe ratio of ≈ 1.45 — **with no skill at all**
- ▣ The Monte Carlo simulation agrees with the theorem

 MFM_ch13_backtest_overfitting

In-Sample vs. Out-of-Sample: Bitcoin Grid Search (1/2)



In-Sample vs. Out-of-Sample: Bitcoin Grid Search (2/2)

- Moving-average (MA) crossover on Bitcoin: long when the fast MA of the log price exceeds the slow MA by more than a band; 1,279 configurations (fast, slow, band); in sample 15 May 2015 – 2020, out of sample 2021–2026
- Selected rule (14/20 days): SR_{IS} **1.89** $\rightarrow$ SR_{OOS} **0.02** (annualised with $\sqrt{365}$); Spearman rank correlation IS–OOS $\rho = 0.18$
- Median OOS SR 0.55 for the top 10% in sample vs. 0.42 for all: the *family* of trend rules carries information, the *selected* optimum is noise

 MFM_ch13_backtest_overfitting

Sharpe Ratio Inference: Standard Error and PSR (1/2)

- ▣ Per-period Sharpe ratio (SR) of daily log returns, risk-free rate 0: $\widehat{SR} = \bar{r}/\hat{\sigma}$, the mean return $\bar{r}$ over its standard deviation $\hat{\sigma}$
- ▣ Shape of the return distribution, with μ , σ the mean and SD of r
 - ▶ skewness $\gamma_3 = \mathbb{E}[(r - \mu)^3]/\sigma^3$: < 0 when large losses are more frequent than large gains
 - ▶ ordinary (not excess) kurtosis $\gamma_4 = \mathbb{E}[(r - \mu)^4]/\sigma^4$: 3 for the Normal distribution, larger for fat tails
- ▣ Sampling variance of $\widehat{SR}$, i.i.d. returns, delta method (Lo (2002); Opdyke (2007)):

$$\text{Var}(\widehat{SR}) \approx \frac{1}{T} \left(1 - \gamma_3 SR + \frac{\gamma_4 - 1}{4} SR^2 \right)$$

- ▶ T : number of returns; SR : true per-period Sharpe ratio
- ▶ negative skewness and fat tails widen the uncertainty; Normal distribution ($\gamma_3 = 0$, $\gamma_4 = 3$): $(1 + \frac{1}{2}SR^2)/T$
- ▣ Annualising with $\sqrt{A}$, A return periods per year (252 trading days; 365 for Bitcoin), needs zero autocorrelation; otherwise $SR_A = \eta(A) SR$ (Lo (2002))
 - ▶ $\eta(A) = A/\sqrt{A + 2 \sum_{k=1}^{A-1} (A-k)\rho_k}$, ρ_k : autocorrelation of returns at lag k ; $\rho_k > 0$ makes $\eta(A) < \sqrt{A}$

Sharpe Ratio Inference: Standard Error and PSR (2/2)

Definition 5 (Probabilistic Sharpe Ratio (PSR), Bailey & López de Prado (2012))

▣ Probability that the true SR exceeds a benchmark SR^* , the variance above in the denominator:

$$\widehat{PSR}(SR^*) = \Phi \left(\frac{(\widehat{SR} - SR^*)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}^2}} \right)$$

- ▣ SR^* : the benchmark, e.g. 0 or the expected maximum SR_0 ; $\hat{\gamma}_3$, $\hat{\gamma}_4$: sample skewness and kurtosis
- ▣ The argument of Φ is a z-statistic: the distance $\widehat{SR} - SR^*$ in standard errors
- ▣ Reading: $\widehat{PSR} > 0.95$: the true SR exceeds SR^* with 95% confidence
- ▣ Two strategies on the same days: test $SR_a - SR_b$ with the HAC delta method or a studentised block bootstrap (Ledoit & Wolf (2008)), not with two separate intervals

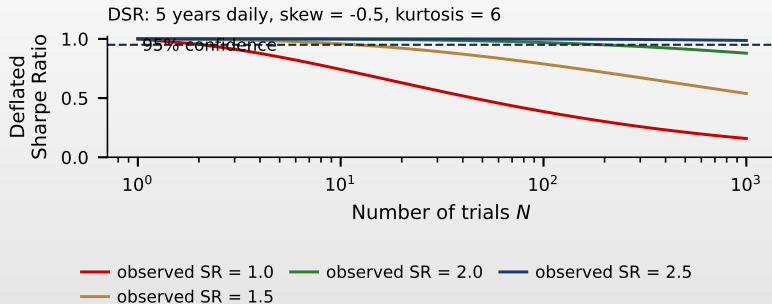
Deflated Sharpe Ratio: the Best Bitcoin Rule

- **Deflated Sharpe Ratio** (DSR, Bailey & López de Prado (2014)): $DSR = \widehat{PSR}(SR_0)$, the PSR with the benchmark set to SR_0 , the expected maximum of the False Strategy Theorem computed with $\text{Var}[\widehat{SR}_n]$ of the trials
 - ▶ corrects for selection (N trials), non-Normality and sample length; reject unless $DSR > 0.95$, and report N
- Worked example, best in-sample rule: $T = 2,058$ days, $\widehat{SR} = 0.0991$ a day (1.89 annualised), $\hat{\gamma}_3 = 0.12$, $\hat{\gamma}_4 = 12.33$
 - ▶ denominator $\sqrt{1 - 0.12 \cdot 0.0991 + \frac{12.33-1}{4} \cdot 0.0991^2} = 1.008$; $N = 1,279$ trials
 - ▶ $z = (\widehat{SR} - SR_0)\sqrt{T-1}/1.008$, the argument of Φ ; $DSR = \Phi(z)$; $SD = \sqrt{\text{Var}[\widehat{SR}_n]}$

Var $[\widehat{SR}_n]$ from	SD	SR_0 (per day)	z / DSR
dispersion of the 1,279 trial SRs	0.0106	0.035	2.88 / 0.998
independent trials, $1/T$	0.0220	0.073	1.16 / 0.877

- The trials are highly correlated, so their dispersion understates the search: the verdict depends on this choice, and the out-of-sample SR of 0.02 sides with the conservative value

Deflated Sharpe Ratio in Practice



- 5 years of daily returns, skewness -0.5 , kurtosis 6
- An annualised SR of 1.0 is significant only if (almost) no other strategy was tried: it fails after 2–3 trials
- SR = 1.5 survives ≈ 10 trials, SR = 2.0 ≈ 200 ; only SR = 2.5 survives 1000

Data-Snooping Tests on the Bitcoin Grid (1/2)

- ▣ K rules; $r_{k,t}$: return of rule k on day t ; $r_{0,t}$: benchmark return; differential $d_{k,t} = r_{k,t} - r_{0,t}$; $\bar{d}_k$: its mean over the n days
- ▣ $H_0 : \max_k \mathbb{E}[d_{k,t}] \leq 0$: no rule beats the benchmark
- ▣ Null distribution from a **stationary bootstrap** (Politis & Romano (1994)): resample blocks of whole days, of random length, the same days for all rules
 - ▶ keeps the correlation between rules: no “effective” N needed
- ▣ Reality Check (RC, White (2000)): statistic $\max_k \sqrt{n} \bar{d}_k$, the scaled mean of the best rule; p -value: share of bootstrap maxima above it
- ▣ Superior Predictive Ability (SPA, Hansen (2005)): studentised, $\max_k \sqrt{n} \bar{d}_k / \hat{\omega}_k$ with $\hat{\omega}_k$ the bootstrap SD of $\sqrt{n} \bar{d}_k$; clearly bad rules not recentred, so adding junk rules does not destroy power
- ▣ Romano–Wolf StepM (Romano & Wolf (2005)): names *which* rules win, with family-wise error rate (FWER) control, the probability of at least one false rejection

Data-Snooping Tests on the Bitcoin Grid (2/2)

- Setup: 1,279 rules, in-sample $n = 2,058$ days, mean block 20 days, 2,000 replications

Benchmark	largest studentised mean	RC p	SPA p	StepM rules kept
cash ($r_0 = 0$)	–	0.004	0.002	314
buy-and-hold	0.88 (the 14/20 rule)	0.74	0.70	0

- Against cash, some trend rule earns a positive mean: Bitcoin's own trend; no rule beats holding Bitcoin, even in sample, consistent with the OOS SR of 0.02
- DSR asks “above the luck of N trials?” and needs N and $\text{Var}[\widehat{SR}]$ (0.998 or 0.877); SPA takes both from the data

Probability of Backtest Overfitting (PBO)

Combinatorially symmetric cross-validation (CSCV): Bailey, Borwein, López de Prado & Zhu (2017)

1. Matrix of returns: T periods $\times$ N configurations; split the rows into S contiguous blocks
 2. For each of the $\binom{S}{S/2}$ choices, half of the blocks are in sample (IS), the rest out of sample (OOS)
 3. Pick the best IS configuration n^* ; rank all N configurations OOS in ascending order and set $\bar{\omega} = \text{rank}(n^*) / (N + 1) \in (0, 1)$
 4. Logit $\lambda = \log \frac{\bar{\omega}}{1 - \bar{\omega}}$; $PBO = \Pr[\lambda \leq 0]$: the share of splits in which the IS winner ranks at or below the OOS median
- $\binom{S}{S/2}$: number of ways to choose $S/2$ of the S blocks; $\bar{\omega}$: relative OOS rank of the IS winner, near 1 if it stays the best; $\lambda > 0$: above the OOS median
 - $PBO \approx 0$: the IS winner reliably stays above the OOS median; $PBO \approx 0.5$ or more: selecting the IS winner is no better than picking at random
 - PBO evaluates the **research procedure**, not one strategy, and is not the probability of losing money; like DSR it needs **every** trial logged (Seminar B12)

Selection Is Not Inference: Post-Double Selection (1/2)

- The LASSO path answers “which feature enters first”, not “which feature matters”; OLS t -statistics after LASSO ignore the selection step
- A control correlated with the regressor of interest d_t but only weakly with y_t gets dropped: its effect moves into the coefficient of d_t (omitted-variable bias)
- Partially linear model: the effect of d_t is linear, the controls enter through unknown functions

$$y_t = \theta d_t + g(x_t) + \varepsilon_t, \quad d_t = m(x_t) + v_t$$

- ▶ y_t : outcome; d_t : regressor of interest; θ : its effect, the parameter to estimate
- ▶ x_t : the p controls; g : their effect on y_t ; m : how they predict d_t ; ε_t , v_t : errors
- ▶ approximately sparse: a few controls explain most of g and m

Selection Is Not Inference: Post-Double Selection (2/2)

- Post-double selection (Belloni, Chernozhukov & Hansen (2014)):
 1. LASSO of y_t on $x_t \Rightarrow$ selected set $\hat{S}_y$
 2. LASSO of d_t on $x_t \Rightarrow$ selected set $\hat{S}_d$
 3. OLS of y_t on d_t and $x_{\hat{S}_y \cup \hat{S}_d}$, robust standard error
- $\hat{S}_y, \hat{S}_d$: indices of the controls kept by each LASSO; $x_{\hat{S}_y \cup \hat{S}_d}$: the controls kept by either
- A control is missed only if it matters for *neither* equation, so the omitted bias is second order
- Valid under independent sampling and sparsity, with HAC errors an approximation for time series (penalty and conditions in the [Appendix](#))

Double/Debiased ML: Orthogonal Scores and Cross-Fitting

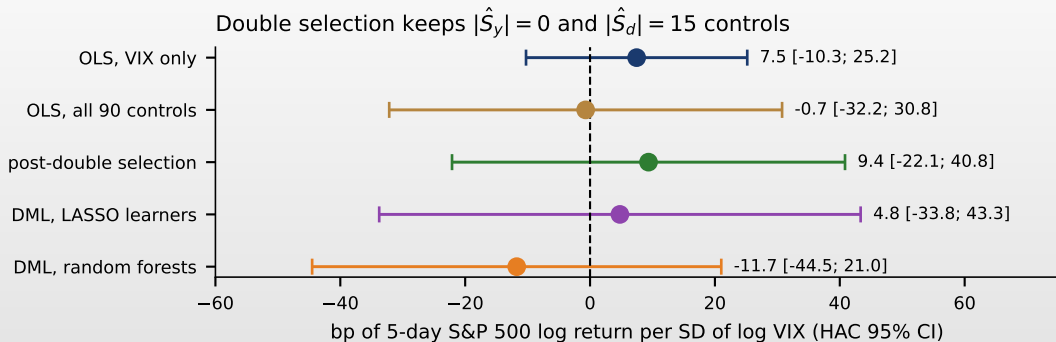
- Chernozhukov et al. (2018): residualise both variables with *any* learner (LASSO, RF, boosting), then regress residual on residual

$$\tilde{y}_t = y_t - \hat{\ell}(x_t), \quad \tilde{v}_t = d_t - \hat{m}(x_t), \quad \hat{\theta} = \frac{\sum_t \tilde{v}_t \tilde{y}_t}{\sum_t \tilde{v}_t^2}$$

- ▶ $\hat{\ell}(x) = \hat{\mathbb{E}}[y \mid x]$: ML prediction of y_t from the controls; $\hat{m}(x)$: ML prediction of d_t
- ▶ $\tilde{y}_t, \tilde{v}_t$: the parts of y_t and d_t the controls do not explain; $\hat{\theta}$: OLS slope of $\tilde{y}_t$ on $\tilde{v}_t$

- Neyman-orthogonal score $\psi = (\tilde{y} - \theta \tilde{v}) \tilde{v}$, whose mean is zero at the true θ
 - ▶ first-order errors in $\hat{\ell}$ and $\hat{m}$ do not bias $\hat{\theta}$ (rate conditions in the [Appendix](#))
- Cross-fitting: fit $\hat{\ell}, \hat{m}$ on $K - 1$ folds, residualise the held-out fold, repeat; removes the own-observation overfitting bias
- The published theory assumes independent sampling; with overlapping 5-day labels we use contiguous folds purged by 5 days around the test fold and HAC standard errors, a data limitation stated as such, not a new estimator

Worked Example: Does the VIX Predict S&P 500 Returns? (1/2)



✗ single LASSO on (VIX, controls): VIX dropped, no standard error --- no predictive effect

Worked Example: Does the VIX Predict S&P 500 Returns? (2/2)

- y_t : 5-day forward log return (bp); d_t : log VIX, standardised (1 SD = 0.356); 90 controls: 12 technical features, squares, pairwise products; $n = 6,269$; HAC errors, 4 lags
- The single LASSO drops the VIX and every control: “no effect”, with no standard error
- Double selection keeps $|\hat{S}_y| = 0$ and $|\hat{S}_d| = 15$ controls ($|\cdot|$: number of elements); 95% CI for θ $[-22.1; 40.8]$ bp per week per SD
- Honest intervals contain zero and large effects of both signs: the data cannot decide

The Virtue of Complexity, and a Critical Reading

- Kelly, Malamud & Zhou (2024): ridge regressions on P random Fourier features of a few predictors, rolling windows as short as $T = 12$ months, P in the thousands
 - ▶ P : number of features (the paper's notation); random Fourier features: sin and cos of random linear combinations of the predictors; T : training window, in months
 - ▶ theory: with $P > T$ the minimum-norm solution (the exact fit with the smallest $\|\beta\|_2$) shrinks implicitly ("double descent"); R_{OOS}^2 may be negative while the Sharpe ratio of market timing **rises** with complexity; complexity is not automatically overfitting
- Nagel (2025): for $P \gg T$ the forecast is a similarity-weighted average of the T training returns
 - ▶ in short windows similarity is mostly recency, so the method becomes a **volatility-timed momentum** strategy, which loses on artificial data with reversals
- Avramov, Cheng & Metzker (2023): deep-learning signal profits come from hard-to-arbitrage stocks
 - ▶ excluding microcaps, distressed firms or high-volatility episodes attenuates them, and costs erode them further
- Chen, Pelger & Zhu (2024): impose economic structure instead; the no-arbitrage moment conditions become the loss of a deep network (next slides)
- Survey of the field: Kelly & Xiu (2023)

Case Study: Deep Learning in Asset Pricing, Question and Method (1/2)

- Chen, Pelger & Zhu (2024): estimate the stochastic discount factor (SDF) M_{t+1} that prices **all individual US stocks**
 - ▶ conditioning on 46 firm characteristics and on the history of 178 macroeconomic series
- No arbitrage: the SDF prices every excess return (Section I.A, eq. 1)

$$E_t[M_{t+1}R_{t+1,i}^e] = 0, \quad M_{t+1} = 1 - \omega_t^\top R_{t+1}^e$$

- ▶ $R_{t+1,i}^e$: excess return of stock i over the Treasury bill from month t to $t+1$; R_{t+1}^e : the vector of all stocks
- ▶ E_t : expectation given the information at t ; ω_t : portfolio weights, a function of characteristics and macro states
- $F_{t+1} = \omega_t^\top R_{t+1}^e$, the SDF portfolio, is the conditional tangency portfolio (highest Sharpe ratio), and

$$E_t[R_{t+1,i}^e] = \beta_{t,i} E_t[F_{t+1}]$$

- ▶ $\beta_{t,i}$: loading of stock i on F ; a stock's expected excess return is its loading times the premium of F

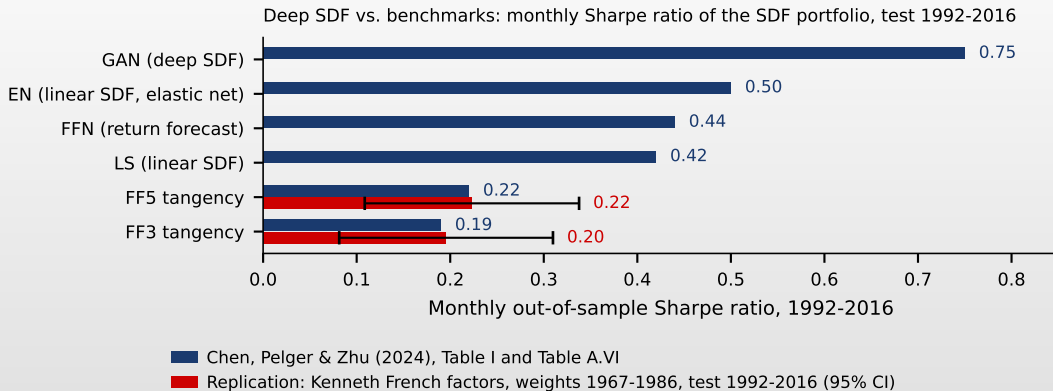
Case Study: Deep Learning in Asset Pricing, Question and Method (2/2)

- Why: forecasting (Gu, Kelly & Xiu (2020), the benchmark feedforward network, FFN) fits a second moment where signal-to-noise is low
 - ▶ the no-arbitrage condition targets a first moment, and averaging over time removes much of the noise
- Method: adversarial generalized method of moments (GMM, eq. 3, after Hansen & Jagannathan (1997))

$$\min_{\omega} \max_g \frac{1}{N} \sum_{j=1}^N \left(E[M_{t+1} R_{t+1,j}^e g(I_t, I_{t,j})] \right)^2$$

- ▶ I_t : macroeconomic information; $I_{t,j}$: characteristics of stock j ; N : number of stocks
- ▶ g : instrument function chosen by an adversary to make the pricing errors as large as possible; ω minimises this worst case
- Networks: an FFN for ω , a long short-term memory network (LSTM) that turns the macro history into 4 states, an adversary g that builds the hardest test assets; the full model is labelled GAN
- Chronological split: training 1967–1986, validation 1987–1991, test 1992–2016; configurations ranked by the validation Sharpe ratio (data and tuning in the [Appendix](#))

Deep Learning in Asset Pricing: Sharpe Ratios, Paper and Replication (1/2)

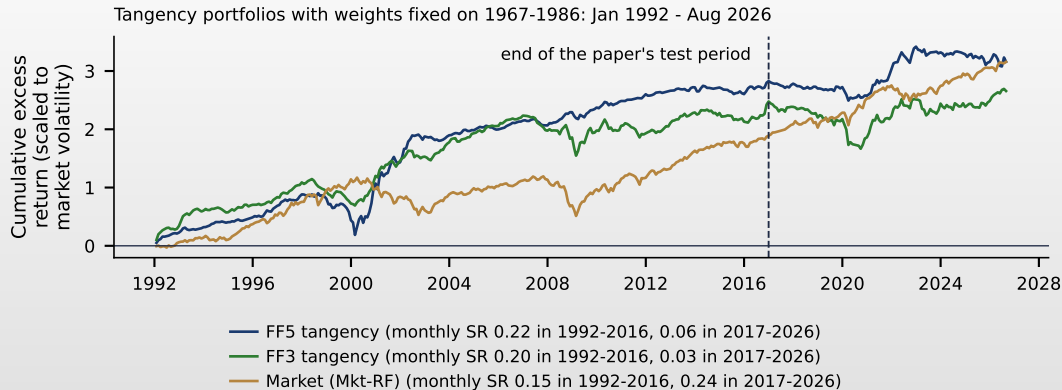


Deep Learning in Asset Pricing: Sharpe Ratios, Paper and Replication (2/2)

- Blue: the paper, monthly out-of-sample SR 1992–2016: GAN 0.75, elastic net linear SDF (EN) 0.50, FFN 0.44, unregularised linear SDF (LS) 0.42 (Table I)
 - ▶ FF5 0.22, FF3 0.19 (Table A.VI)
 - ▶ the regularised no-arbitrage model already beats the forecasting network
- Red: FF3 and FF5 tangency portfolios $w \propto \hat{\Sigma}^{-1} \hat{\mu}$ on the [Kenneth French](#) factors, weights fixed on 1967–1986: FF5 0.22 [0.11; 0.34], FF3 0.20 [0.08; 0.31] (SE 0.058, [Lo \(2002\)](#)): both published values lie inside the intervals
 - ▶ $\hat{\mu}$, $\hat{\Sigma}$: sample mean vector and covariance matrix of the factor returns; $\propto$: proportional to

 MFM_ch13_sdf_replication

Deep Learning in Asset Pricing: the Benchmarks after 2016 (1/2)



Deep Learning in Asset Pricing: the Benchmarks after 2016 (2/2)

- Same fixed weights, January 1992 – August 2026, each series scaled to the volatility of the market excess return on 1992–2016
 - ▶ monthly SR 2017–2026 ($T = 116$, $SE \approx 0.09$): FF5 0.06, FF3 0.03, market 0.24; on 1992–2016: 0.22, 0.20, 0.15
- Replicable with public data: the Fama–French benchmarks
 - ▶ not replicable: the stock-level GAN, LS and EN (the CRSP panel is not in the course data)
 - ▶ the post-2016 fall is a warning, not yet a test, and the same question applies to a GAN frozen in 1991 (details in the [Appendix](#))

 MFM_ch13_sdf_replication

An Open Question: Discovery or Data Mining at Scale?

- Chen, Lopez-Lira & Zimmermann (2022): mining 29,000 accounting ratios for $t > 2$ gives cross-sectional predictability similar to peer-reviewed predictors
 - ▶ for both, about half remains after the original samples
- Novy-Marx & Velikov (2025): an LLM (large language model) pipeline mines over 30,000 signals, keeps 96 that pass a fixed protocol
 - ▶ writes hundreds of complete papers, with invented motivations and some imagined citations
 - ▶ HARKing (hypothesising after the results are known) at scale
- Open: when a machine proposes features, model and story
 - ▶ what is the number of trials?
 - ▶ what separates a discovery from the best of many draws?
- The discovery loop: literature → competing hypotheses → data and code → robustness fixed in advance → referee-style critique → report every test (reusable prompts in the [▶ Appendix](#))
- What the human checks: every reference resolves (DOI, arXiv); every trial is logged, so M in the correction counts them all; correlated tests get a joint null; transformations live inside the training fold

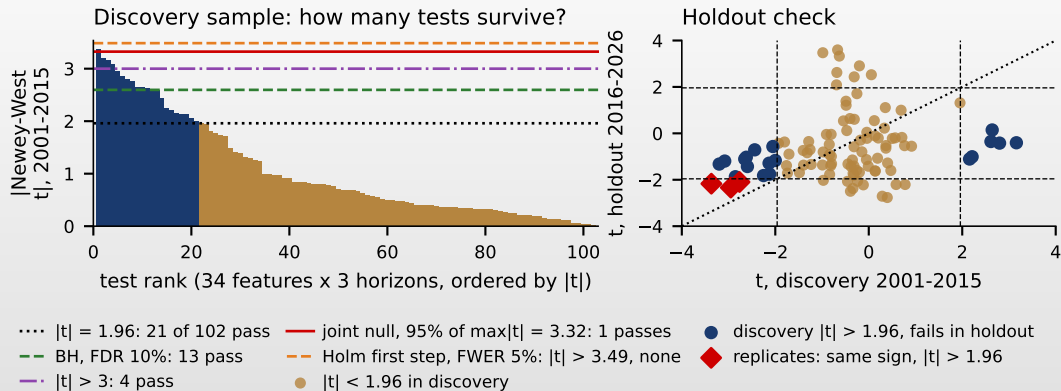
Mini-Case: 102 Tests, One Old Survivor (1/2)

- A typical assistant proposal: “add more indicators: momentum, moving-average distances, volatility ratios, RSI and VIX changes over several windows”
- 34 features $\times$ 3 horizons (1, 5, 20 days) = 102 predictive regressions on 2001–2015, Newey–West t
 - ▶ regression j : $r_{t,t+h} = \alpha_j + \beta_j x_{j,t} + e_t$, with $x_{j,t}$ feature j and $r_{t,t+h}$ the next h -day return; $H_{0,j} : \beta_j = 0$
- Corrections for $M = 102$ tests, $p_{(1)} \leq \dots \leq p_{(M)}$ the ordered p -values
 - ▶ Holm: reject while $p_{(j)} \leq 0.05/(M - j + 1)$; controls the FWER
 - ▶ Benjamini–Hochberg (BH, [Benjamini & Hochberg \(1995\)](#)): reject the j smallest, for the largest j with $p_{(j)} \leq j q / M$
 - ▶ controls the false discovery rate (FDR) at q , the expected share of false rejections among all rejections
 - ▶ joint null: the 95% quantile of $\max_j |t_j|$ when the target is shifted circularly, which keeps the correlation between the tests

Mini-Case: 102 Tests, One Old Survivor (2/2)

- ▣ $|t| > 1.96$: **21** of 102, against 5.1 expected under the global null at 5%
- ▣ Holm (FWER 5%): **0**; BH (FDR 10%): **13**; $|t| > 3$ (Harvey, Liu & Zhu (2016)): **4**
- ▣ Joint null (500 circular shifts of the target): 95% quantile of $\max_j |t_j| = 3.32$
 - ▶ one survivor, the 1-day return on the next day's return, $t = -3.37$: short-term reversal, a known effect
- ▣ Holdout 2016–2026: 3 of the 21 keep their sign with $|t| > 1.96$, all reversal variants; the VIX-based “finds” fail
- ▣ Holm and BH control different error rates at different levels; BH needs independence or positive dependence of the true-null p-values, which correlated predictors do not ensure

Mini-Case: Thresholds and the Holdout (1/2)



Mini-Case: Thresholds and the Holdout (2/2)

- Left: the 102 discovery $|t|$ in decreasing order; each threshold answers a different question, and only the joint null accounts for the correlation between the tests
- Right: discovery vs. holdout t ; the three replications (red) are all short-term reversal; the significant VIX features collapse towards zero

 MFM_ch13_ai_discovery

Your Turn: a Project Seed

▣ Questions

- ▶ can an AI-assisted feature search find a predictor of index returns (S&P 500, BET, Bitcoin)?
- ▶ does it survive a pre-registered correction and a holdout?
- ▶ does it go beyond short-term reversal?

▣ Replicate first: this mini-case (21 of 102, one survivor at the joint null)

▣ Deliverables

- ▶ a feature list fixed before the data are touched
- ▶ all tests with Holm, BH and the max- $|t|$ null; a holdout
- ▶ DSR and PBO for any strategy built on the survivors

▣ Log ([templates](#))

- ▶ each prompt that shaped a decision in AI_USE.md
- ▶ each invented reference, leak or undercounted trial in AI_ERRORS.md, with the test that caught it

Key Takeaways and a Decision Checklist

Three empirical lessons

- An ADF rejection is not stationarity: truncated FFD of the S&P 500 log price stays $I(1)$; estimate memory with local Whittle
- Shuffled K -fold reports 69% on pure noise; on the S&P 500 no model beats “always up” (PT, DM, GW)
- The best of many backtests must be deflated or bootstrapped: Bitcoin grid 1.89 in sample, 0.02 out of sample

Next

- [Chapter 14](#): deep learning and time-series foundation models; [Chapter 15](#): large language models and sentiment

Before any claim

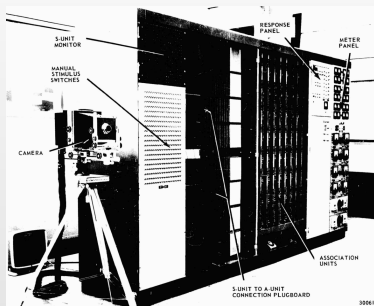
- Point-in-time data; transformations fitted on training folds only
- Volatility-scaled labels with recorded spans
- Purged K -fold or CPCV with embargo; walk-forward for the final test
- A naive baseline; PT, DM / CW / GW and block-bootstrap intervals
- Costs included; every trial logged; DSR, SPA and PBO
- “Selected” or “important” is not “significant”: double selection or DML for inference

Appendix: Further Study on Quantinar

Advanced courses on quantinar.com

- ☐  Random Forests
- ☐  Shapley Values
- ☐  Shapley Curves: A Smoothing Perspective
- ☐  Machine Learning in Financial Risk
- ☐  Introduction to Reinforcement Learning
- ☐  Generative Adversarial Networks
- ☐  MSCA Digital: The Need for Explainable AI
- ☐  Business Analytics and Data Science
- ☐  Model Selection Criteria
- ☐  PAC Learning

Appendix: From the Perceptron to Deep Blue



Mark I Perceptron, operator's manual (1960)

Figure: J.C. Hay, A.E. Murray (1960); public domain; Wikimedia Commons



IBM Deep Blue, Computer History Museum

Photo: Anton Chiang (2011); CC BY 2.0; Wikimedia Commons

- **Perceptron Rosenblatt (1958)**
 - ▶ a learning rule for a single artificial neuron
 - ▶ the Mark I: hardware built to recognise simple images
- 1997: IBM's Deep Blue beats world chess champion Garry Kasparov [Campbell, Hoane & Hsu \(2002\)](#)
- Deep Blue searched with hand-built rules; modern ML (this chapter) *learns* the rules from data

Appendix: the In-Sample R^2 of a Predictor without Skill

- ▣ A monthly R^2 of 0.5% is economically large, but sampling noise is of the same order
- ▣ In-sample OLS R^2 of one predictor with no skill (intercept, independent errors, $n \approx 300$ months): $nR^2 \approx \chi_1^2$
 - ▶ n : number of months; χ_1^2 : chi-squared distribution with 1 degree of freedom, mean 1 and variance 2
 - ▶ so R^2 has mean $1/n \approx 0.3\%$ and standard deviation $\sqrt{2}/n \approx 0.5\%$
- ▣ R_{OOS}^2 can be negative, has no such null distribution and needs its own test (Clark–West)

Appendix: Worked Example: How a Forest Averages Its Trees

- ▣ Random forest of the case study (400 trees, minimum leaf size 100), trained on the first four purged folds
- ▣ Prediction for the last observation (11 September 2026): $P(\text{up in 5 days})$ from each tree
 - ▶ individual trees range from 0.31 to 0.88 (standard deviation 0.11)
 - ▶ 81.5% of the trees vote “up” ($p > 0.5$)
 - ▶ forest probability = average of the 400 trees = 0.608
- ▣ Single trees disagree strongly; the average is far more stable
- ▣ Stability is not accuracy: the forest still does not beat “always up” (case study)

Appendix: Sampling Financial Data: Bars

Time and tick bars

- ▣ A **bar** = one row (open, high, low, close, volume) summarising a sampling interval
- ▣ Time bars: every fixed interval (1 minute, 1 day); oversample quiet periods; fat-tailed returns
- ▣ Tick bars: one bar every N transactions

Volume, dollar and information bars

- ▣ One bar every V shares or every D dollars traded: sampling by **information arrival**
- ▣ Returns closer to i.i.d. draws from the Normal distribution: [Clark \(1973\)](#); [Ané & Geman \(2000\)](#)
- ▣ Imbalance / runs bars: sample when order flow becomes one-sided

- ▣ This chapter uses daily time bars; tick data (crypto exchanges) in [Chapters 10 and 16](#)

Appendix: Fractional Weights and Their Asymptotics

Definition 6 (Fractional difference operator)

- With the backshift operator $BX_t = X_{t-1}$ and a real order d :

$$(1 - B)^d = \sum_{k=0}^{\infty} \binom{d}{k} (-B)^k, \quad \tilde{X}_t = \sum_{k=0}^{\infty} w_k X_{t-k}$$

- Weights: $w_0 = 1$, $w_k = -w_{k-1} \frac{d-k+1}{k} = \frac{\Gamma(k-d)}{\Gamma(-d)\Gamma(k+1)} \sim \frac{k^{-(1+d)}}{\Gamma(-d)}$ as $k \rightarrow \infty$
- $\binom{d}{k} = d(d-1)\cdots(d-k+1)/k!$: the binomial coefficient, defined for any real d ; Γ : the gamma function, $\Gamma(k+1) = k!$
- $\sim$: the ratio of the two sides tends to 1; the weights decay like $k^{-(1+d)}$, more slowly for smaller d
- Infinite sum: $\sum_{k=0}^{\infty} w_k = (1-1)^d = 0$ follows from the binomial series at $B = 1$

Appendix: Local Whittle and Exact Local Whittle (1/2)

- Periodogram: the sample estimate of the spectral density at the Fourier frequency $\lambda_j = 2\pi j/n$

$$I(\lambda_j) = \frac{1}{2\pi n} \left| \sum_{t=1}^n X_t e^{-i\lambda_j t} \right|^2$$

- Local Whittle ([Robinson \(1995\)](#)): choose δ to fit $f(\lambda) \approx G\lambda^{-2\delta}$ on the first m frequencies

$$\hat{\delta} = \arg \min_{\delta} \log \left(\frac{1}{m} \sum_{j=1}^m \lambda_j^{2\delta} I(\lambda_j) \right) - \frac{2\delta}{m} \sum_{j=1}^m \log \lambda_j, \quad \sqrt{m}(\hat{\delta} - \delta) \xrightarrow{d} N(0, \frac{1}{4})$$

- n : sample size; m : number of frequencies used; i : the imaginary unit; G has been profiled out
- $\xrightarrow{d}$: convergence in distribution; hence the standard error $1/(2\sqrt{m})$

Appendix: Local Whittle and Exact Local Whittle (2/2)

- Exact local Whittle (ELW, [Shimotsu & Phillips \(2005\)](#); unknown mean: [Shimotsu \(2010\)](#)): periodogram of $(1 - B)^\delta X_t$, valid also for non-stationary $\delta \geq \frac{1}{2}$
- Why the FFD series looks like $\hat{\delta} \approx 0.75$: above frequency $1/K$ the truncated filter acts as $(1 - B)^{0.25}$; only $n/(2\pi K) \approx 2$ of the $m = 294$ frequencies lie below $1/K$, where the unit root returns
 - ▶ K : the FFD window length (444 lags)
 - ▶ the estimate is the apparent memory over the band, not the integration order of the series

 MFM_ch13_ffd

Appendix: Worked Example: Purging with 10 Observations

- Observations $t = 0, \dots, 9$; label of t ends at $t + 2$ (capped at 9)
- 5 contiguous folds of 2; embargo = 1 observation

Test	Removed	Train
0, 1	2, 3	4–9
2, 3	0, 1, 4, 5	6–9
4, 5	2, 3, 6, 7	0, 1, 8, 9
6, 7	4, 5, 8, 9	0–3
8, 9	6, 7	0–5

- Test $\{4, 5\}$: labels of 2 and 3 end at 4 and 5 $\Rightarrow$ purged; 6 is the embargo; 7 starts when the last test label ends $\Rightarrow$ purged
- With overlapping labels, up to 4 of 10 observations leave the training set

Appendix: When Is K-Fold Valid for Time Series?

- Bergmeir, Hyndman & Koo (2018): for a **purely autoregressive** model $y_t = g(y_{t-1}, \dots, y_{t-p}) + \varepsilon_t$ with serially uncorrelated errors, standard K -fold CV is valid
 - ▶ g : any, possibly non-linear, function of the last p values; ε_t : error
 - ▶ the errors in the test fold are uncorrelated with the training errors, so the CV error is unbiased for the OOS error
- It fails when the **errors** are serially correlated
 - ▶ overlapping labels: h -day forward returns share $h - 1$ daily returns
 - ▶ persistent features that are not in the model's lag set (rolling means, levels such as the FFD log-price)
 - ▶ under-specified dynamics (too few lags)
- The leakage experiment isolates exactly this: 20-day labels and 20-day rolling features
- Check before choosing: the autocorrelation of the CV residuals; if it is significant, purge and embargo

Appendix: Walk-Forward and Combinatorial Purged CV

Walk-forward (WF)

- ▣ Defined on the model-selection slide; mimics real-time deployment; no future data in training
- ▣ But a **single path**: high variance, easy to overfit by repeated trials

Combinatorial purged CV (CPCV)

- ▣ N contiguous groups; test on all $\binom{N}{k}$ combinations of k groups, with purging and embargo
- ▣ $\varphi = \frac{k}{N} \binom{N}{k}$: the number of complete backtest paths; e.g. $N = 6$, $k = 2$: 15 splits, 5 paths
- ▣ A **distribution** of Sharpe ratios instead of one number

- ▣ $\binom{N}{k}$: number of ways to choose k of N groups; each group is tested φ times, so the test predictions form φ full histories

Appendix: Testing Directional Accuracy: Pesaran–Timmermann

- H_0 : the predicted sign x_t is independent of the realised sign y_t (Pesaran & Timmermann (1992)); $\hat{P}$, $\hat{P}^*$, $\hat{p}_y$, $\hat{p}_x$ as on the main slide, n : number of forecasts

$$PT = \frac{\hat{P} - \hat{P}^*}{\sqrt{\hat{V}(\hat{P}) - \hat{V}(\hat{P}^*)}} \xrightarrow{d} N(0, 1), \quad \hat{V}(\hat{P}) = \frac{\hat{P}^*(1 - \hat{P}^*)}{n}$$

- ▶ $\hat{V}(\hat{P}^*) = [(2\hat{p}_y - 1)^2 \hat{p}_x(1 - \hat{p}_x) + (2\hat{p}_x - 1)^2 \hat{p}_y(1 - \hat{p}_y)]/n + 4\hat{p}_y \hat{p}_x(1 - \hat{p}_y)(1 - \hat{p}_x)/n^2$
- ▶ $\hat{V}(\cdot)$: estimated variance under H_0 ; $\hat{P}^*$ is itself estimated, hence the second term
- PT conditions on the marginal frequencies automatically: no need to guess the baseline
- Logit of the case study: $\hat{p}_y = 0.581$, $\hat{p}_x = 0.983$; one-sided $p = 0.90$ for $PT = -1.28$

Appendix: ROC AUC and Its Sampling Uncertainty

- Receiver operating characteristic (ROC) curve: true positive rate $TP/(TP + FN)$ against false positive rate $FP/(FP + TN)$ over all thresholds
- Area under the curve (AUC) = $P(\hat{\rho}_{\text{pos}} > \hat{\rho}_{\text{neg}}) + \frac{1}{2}P(\hat{\rho}_{\text{pos}} = \hat{\rho}_{\text{neg}}) = U/(n_1 n_0)$
 - ▶ $\hat{\rho}_{\text{pos}}, \hat{\rho}_{\text{neg}}$: predicted probabilities of a randomly drawn positive (up) and negative (down) observation
 - ▶ n_1, n_0 : numbers of positives and negatives; U : Mann–Whitney statistic, the number of (positive, negative) pairs ranked correctly, ties counted $\frac{1}{2}$
 - ▶ 0.5 = no ranking ability; 1 = every positive ranked above every negative
- DeLong standard error (DeLong, DeLong & Clarke-Pearson (1988)): from the placement values, the share of the other class that each observation outranks; assumes independent observations
- Logit, pooled out-of-fold AUC = 0.479
 - ▶ DeLong 95% CI [0.465; 0.494]: “significantly worse than random”
 - ▶ block bootstrap 95% CI [0.455; 0.504]: noise around 0.5
- An AUC below 0.5 is anti-ranking only if it survives a test that respects the dependence

Appendix: Giacomini–White, Statistic and Rolling Window

- **Giacomini & White (2006)**: the null concerns the forecasting **method** (model + estimation window + refitting rule), not the population model

$$H_0 : \mathbb{E}[d_{t+\tau} \mid \mathcal{F}_t] = 0 \Rightarrow \mathbb{E}[h_t d_{t+\tau}] = 0, \quad GW = n \bar{Z}^\top \hat{\Omega}^{-1} \bar{Z} \xrightarrow{d} \chi_q^2$$

- Notation

- ▶ $d_{t+\tau}$: loss differential of the two forecasts made at t for horizon $\tau = 5$ days; $\mathcal{F}_t$: information available at t
- ▶ h_t : test functions known at t (here 1 and the lagged loss differential, $q = 2$); $Z_t = h_t d_{t+\tau}$
- ▶ $\bar{Z}$: mean of Z_t over the n forecasts; $\hat{\Omega}$: HAC covariance matrix of Z_t ; χ_q^2 : chi-squared with q degrees of freedom

- Reading: under H_0 no information at t predicts which method will do better; a large GW (small p) rejects equal conditional accuracy
- The theory requires a bounded (rolling) estimation window, so that estimation noise never vanishes; an expanding window lies outside the theorem
- S&P 500 logit, rolling 8-year window: $GW = 26.5$ ($p < 0.001$): the logit forecasts worse than the constant

 MFM_ch13_inference

Appendix: Feature Importance: SHAP

Definition 7 (Shapley value of feature j (SHAP, Lundberg & Lee (2017)))

- Average marginal contribution of j over all coalitions $S \subseteq \{1, \dots, p\} \setminus \{j\}$:

$$\phi_j = \sum_S \frac{|S|! (p - |S| - 1)!}{p!} [v(S \cup \{j\}) - v(S)], \quad \hat{f}(\mathbf{x}) = \phi_{\mathbf{o}} + \sum_j \phi_j$$

- Interventional SHAP: $v(S) = \mathbb{E}[\hat{f}(\mathbf{x}_S, \mathbf{X}_{\bar{S}})]$, averaging $\hat{f}$ with the features outside S drawn from a background sample (as in the Quantlet); $\phi_{\mathbf{o}} = v(\emptyset) =$ average prediction

- Notation: p : number of features; S : a coalition, a subset of the other features, $|S|$ its size, $\bar{S}$ the features outside it; $v(S)$: prediction when only the features in S are known; $\hat{f}(\mathbf{x})$: model prediction for $\mathbf{x}$
- The weight of S is the probability that exactly the features in S come before j in a random ordering of the p features
- Conditional SHAP uses $v(S) = \mathbb{E}[\hat{f}(\mathbf{X}) \mid \mathbf{X}_S = \mathbf{x}_S]$; the two differ when features are dependent

 Shapley Values

Appendix: Reading Feature Importances

- A negative MDA is a sign of overfitting, not a proof: sampling noise and correlated features also move MDA
- Remedies for correlated features: permute within clusters, or refit without the feature
- Importances are estimates and need their own uncertainty analysis; honest trees give asymptotically Normal forest *predictions* and CIs ([Wager & Athey \(2018\)](#)), not CIs for importances
- **Backtesting is not a research tool**, importance analysis is: [López de Prado \(2018\)](#)

Appendix: Double Selection and DML, Penalty and Conditions (1/2)

- Partially linear model: $\mathbb{E}[\varepsilon_t \mid d_t, x_t] = 0$, $\mathbb{E}[v_t \mid x_t] = 0$, $\text{Var}(v_t) > 0$; g , m approximately sparse
 - ▶ the errors are unpredictable from the regressors; d_t has variation of its own beyond the controls
- Post-double selection: the heteroskedasticity-robust standard error is valid under independent sampling, regularity conditions and sparsity $s^2 \log^2(p \vee n)/n \rightarrow 0$ (Belloni, Chernozhukov & Hansen (2014))
 - ▶ with time series, HAC errors are an approximation
 - ▶ s : number of controls that matter; p : number of controls; $p \vee n = \max(p, n)$
- LASSO objective, with feature-specific penalty loadings

$$\frac{1}{n} \sum_t (y_t - x_t^\top \beta)^2 + \frac{\lambda}{n} \sum_j \hat{\psi}_j |\beta_j|, \quad \lambda = 2c\sqrt{n} \Phi^{-1}(1 - \gamma/(2p))$$

- ▶ plug-in constants $c = 1.1$, $\gamma = 0.1/\log n$; loadings $\hat{\psi}_j^2 = \frac{1}{n} \sum_t x_{tj}^2 \hat{e}_t^2$, $\hat{e}_t$ the residuals of the previous step, iterated

Appendix: Double Selection and DML, Penalty and Conditions (2/2)

- DML (Chernozhukov et al. (2018)): with $\ell(x) = \mathbb{E}[y | x]$, it suffices (with consistency and regularity) that

$$\|\hat{m} - m\| (\|\hat{\ell} - \ell\| + \|\hat{m} - m\|) = o_P(n^{-1/2})$$

- ▶ $\|\hat{m} - m\|$: root mean squared estimation error of the learner for m ; same for ℓ
- ▶ $o_P(n^{-1/2})$: tends to zero in probability faster than $1/\sqrt{n}$
- ▶ e.g. both errors $o_P(n^{-1/4})$: each learner may converge more slowly than a parametric model, because only the product of the errors matters

Appendix: Machine Learning for Inference in Asset Pricing

- [Feng, Giglio & Xiu \(2020\)](#): does a new factor add pricing information given hundreds of existing ones?
 - ▶ double selection of the controlling factors from the zoo, then a valid t -test of the new factor's risk price
 - ▶ the inferential answer to [Harvey, Liu & Zhu \(2016\)](#), who raise the hurdle to $t > 3$
- Heterogeneous effects: honest causal forests give CIs for effects that vary with covariates ([Wager & Athey \(2018\)](#))
- Three rules
 - ▶ prediction and inference are different targets: a good predictor can give a biased $\hat{\theta}$
 - ▶ orthogonalise before you test, and cross-fit with folds that respect time
 - ▶ report CIs; “selected” or “important” is not “significant”

Appendix: Deep Learning for Sequences

Recurrent networks (LSTM, GRU)

- ▣ Hidden state carries memory: $h_t = g(h_{t-1}, x_t)$
 - ▶ g : a non-linear function with trained weights, x_t : input at t
- ▣ Gates control what is kept or forgotten, mitigating vanishing gradients
- ▣ Used for volatility, limit-order-book and intraday forecasting

Transformers

- ▣ Self-attention: $\text{softmax}(QK^\top / \sqrt{d_k}) V$
 - ▶ query, key, value matrices Q, K, V (linear maps of the inputs)
 - ▶ d_k : key dimension
 - ▶ softmax turns each row of scores into weights that sum to 1
- ▣ Long-range dependence, parallel training
- ▣ Backbone of large language models ([Chapter 15](#))

Appendix: Time-Series Foundation Models

Pre-trained forecasters

- Pre-trained on large collections of series, used **zero-shot** (without re-training): Chronos (Ansari et al., 2024), TimesFM (Das et al., 2024), Moirai (Woo et al., 2024), Lag-Llama (Rasul et al., 2023)
 - Useful baselines for volatility and volume forecasting
 - For returns they inherit the same signal-to-noise limits
 - Pre-training data may overlap your test period: a new form of leakage
-
- Covered in depth, with the risk-forecast evaluation tools of [Chapter 8](#), in [Chapter 14](#)

Appendix: Reinforcement Learning, Generative Models, Explainable AI, Risk

Reinforcement learning (RL)

- A policy $\pi(a | s)$ chooses actions a in states s to maximise cumulative reward
- Optimal execution, market making, deep hedging: Buehler et al. (2019)

Generative models

- GANs, variational autoencoders (VAEs), diffusion models: synthetic scenarios for stress tests

Explainable AI

- Required by regulators (EU AI Act, credit scoring, model risk)
- SHAP, LIME (local interpretable model-agnostic explanations), partial dependence

ML for risk

- Value-at-Risk (VaR): conditional quantiles (quantile forests, quantile-loss boosting)
- Expected Shortfall (ES): average of fitted tail quantiles or a joint VaR-ES loss
- Volatility: a squared-return target (Chapters 7–8)

□ Quantinar:  Reinforcement Learning  GANs  Explainable AI  ML in Financial Risk

◀ Back

Appendix: Deep Learning in Asset Pricing, Data (Section III.A)

- Monthly returns of all CRSP stocks, January 1967 – December 2016; excess over the one-month Treasury-bill rate
- 46 firm characteristics in six groups: past returns, investment, profitability, intangibles, value, trading frictions (Table A.II)
 - ▶ Only stock-months with all 46 available: about 10,000 of 31,000 stocks; ranked each month into cross-sectional quantiles
- 178 macroeconomic series: 124 of FRED-MD ([McCracken & Ng \(2016\)](#)), the 46 monthly medians of the characteristics, 8 predictors of [Welch & Goyal \(2008\)](#); transformed to stationarity

Appendix: Deep Learning in Asset Pricing, Method (Sections I–II, Appendix C) (1/2)

- Objective (eq. 3) and networks: main slide; the adversary g uses 8 instruments; three steps: unconditional SDF, maximise over g , minimise again (Section II.D)
- Tuning: 384 configurations ranked by the validation Sharpe ratio (SR); chosen: 2 layers of 64 units, dropout, Adam; ensemble of 9 fits (Table A.VIII); loadings: a second FFN fits $E_t[R_{t+1,i}^e | F_{t+1}] \propto \beta_{t,i}$
- Benchmarks: the FFN of [Gu, Kelly & Xiu \(2020\)](#); the linear SDF $\omega_{t,i} = \theta^\top l_{t,i}$ on long and short characteristic legs, unregularised (LS) or elastic net (EN); FF3/FF5 tangency portfolios ([Fama & French \(2015\)](#))
 - ▶ $\omega_{t,i}$: weight of stock i ; $l_{t,i}$: its characteristics; θ : one coefficient per characteristic, the same for all stocks and months

Appendix: Deep Learning in Asset Pricing, Method (Sections I–II, Appendix C) (2/2)

- ▣ Metrics (II.F): the SR of F and two fit measures; $\hat{\epsilon}_{t+1,i}$: pricing error, the part of $R_{t+1,i}^e$ not explained by $\hat{\beta}_{t,i}F_{t+1}$
- ▣ Explained variation: one minus the average squared pricing error over the average squared excess return

$$EV = 1 - \frac{\overline{\hat{\epsilon}^2}}{(\overline{R^e})^2}$$

- ▶ overline: time average of the monthly cross-sectional means; $EV = 1$: the factor explains all variation
- ▣ Cross-sectional R^2 : how well the model explains average returns

$$XS-R^2 = 1 - \frac{\frac{1}{N} \sum_i \frac{T_i}{T} (\bar{\hat{\epsilon}}_i)^2}{\frac{1}{N} \sum_i \frac{T_i}{T} (\bar{R}_i^e)^2}$$

- ▶ $\bar{\hat{\epsilon}}_i$, $\bar{R}_i^e$: time averages for stock i ; T_i : months of data on stock i , T : months in the sample

Appendix: Deep Learning in Asset Pricing, Results (1/2)

- Source: Chen, Pelger & Zhu (2024), *Management Science*
- Out-of-sample SR of the SDF (Table I), annualised: GAN 2.6, EN 1.7, FFN 1.5 (monthly values on the main slide)
- Individual stocks (Table I): EV 8% (GAN) vs. 3–4%; cross-sectional R^2 23% vs. 14–19%
 - ▶ Reading: the no-arbitrage objective improves both fits, although it maximises neither
- Deciles sorted on the loadings β (Table II, test period, annualised): top minus bottom 0.39; FF5 alpha 0.39, $t = 9.96$; the GRS test rejects the CAPM, FF3 and FF5 ($p = 0.00$)
 - ▶ alpha: average return not explained by the factor model; GRS (Gibbons–Ross–Shanken) test: H_0 all alphas of the decile portfolios are zero
 - ▶ Reading: loadings predict returns, linearly ($R^2 = 0.97$ of mean return on β , Figure 8), and the factor models do not price them

Appendix: Deep Learning in Asset Pricing, Results (2/2)

- 46 sets of characteristic-sorted deciles (Table III): GAN cross-sectional R^2 above 90% for all; EV short-term reversal 0.70 vs. 0.43 (EN), momentum 0.54 vs. 0.26
 - ▶ Reading: gains come from extreme deciles and interactions; size and book-to-market deciles are easy for every model (Section III.E)
- Macro states (Figure 6): the 178 raw series make the out-of-sample SR collapse; GAN loses about 10% without macro states and 20% with unconditional moments only
 - ▶ Reading: the dynamics of macro data must be modelled (LSTM), not fed as the last increment
- Largest stocks (Table IV, monthly SR, models fitted on all stocks, evaluated on the subset)
 - ▶ about 1,500 stocks GAN 0.41, FFN 0.30, EN 0.15
 - ▶ about 550 stocks 0.26, 0.24, 0.23
 - ▶ re-estimated on the 1,500: 0.42, 0.28, 0.42
 - ▶ Reading: the edge narrows sharply among the largest stocks; EN catches up when re-estimated on them

Appendix: Deep Learning in Asset Pricing, Replication and Extension

- With the [Kenneth French Data Library](#) and FRED-MD, on the published split
 - ▶ Replicable: FF3 and FF5 tangency weights estimated on 1967–1986 (footnote 23), monthly SR on 1992–2016 next to Table A.VI
 - ▶ Not replicable with the course data: the published LS and EN (Tables I and III) use rank-weighted legs of 46 characteristics and stock-level betas
 - ▶ Analogue on portfolio data, an extension rather than a replication
 - ▶ the linear SDF ($\hat{\theta}_{LS}$ in closed form, $\hat{\theta}_{EN}$ with the objective of Section II.F) on halves (deciles 1–5, 6–10) of the 15 characteristics that the library sorts into deciles
 - ▶ its levels are not comparable with Tables I and III

Appendix: Deep Learning in Asset Pricing, Limits and an Open Extension

- Limitations stated by the authors
 - ▶ The positivity of the SDF is not imposed (footnote 9); only stock-months with all characteristics, which mainly drops very small stocks (Section III.A)
 - ▶ Frictions (Section III.I, Figure 17): annual SR 1.73 without the smallest 40% of stocks
 - ▶ 2.07 without the widest 40% bid-ask spreads
 - ▶ 1.87 without the lowest 40% turnover; see also [Avramov, Cheng & Metzker \(2023\)](#)
 - ▶ In-sample metrics exceed the out-of-sample ones, a sign of some overfitting (Section III.C); the hidden states are not comparable across fits, so they cannot be averaged (Section II.E)
- Open extension: freeze the 1967–1991 design and evaluate 2017–2026, after the paper's sample
 - ▶ does the linear no-arbitrage SDF still beat FF5?
 - ▶ do LSTM states of FRED-MD add to it?

Appendix: Deep Learning in Asset Pricing, Reading the Two Charts

- GAN 0.75 vs. FF5 0.22
 - ▶ with Lo SEs 0.065 and 0.058, the SE of the difference is at most 0.12 whatever their correlation
 - ▶ the gap exceeds 4 SEs (i.i.d. Normal returns)
 - ▶ a paired test on aligned returns ([Ledoit & Wolf \(2008\)](#)) is sharper
- The fixed FF5 weights load mainly on profitability and investment: RMW 0.42, CMA 0.30 (weights scaled to $\sum |w| = 1$); FF3 puts 0.61 on HML; both combinations were chosen on 1967–1986

 MFM_ch13_sdf_replication

Appendix: The Discovery Loop, Step by Step (1/2)

- **Literature:** "Summarise the evidence on how much published return predictability survives out of sample and after publication: DOI or arXiv id, sample, headline decay estimate."
 - ▶ Resolve every identifier; Semantic Scholar, Google Scholar, Elicit for the search
- **Hypotheses:** "For a feature that predicts S&P 500 returns in 2001-2015, give competing explanations (risk premium, reversal, microstructure, data mining) and what each predicts for 2016-2026 and for other indices."
- **Data and code:** "Write Python that builds 34 technical features from the course's daily S&P 500 and VIX closes with information up to day t only, and reports Newey-West t -statistics for 1-, 5- and 20-day returns in 2001-2015."
 - ▶ Any assistant (Claude, ChatGPT, Gemini, GitHub Copilot); first reproduce the case study: "always up" 58.1%

Appendix: The Discovery Loop, Step by Step (2/2)

- **Robustness**, fixed in advance: "Write the pre-analysis plan: the full list of features and horizons, the holdout, the corrections (Holm, BH, max-t) and the result that would count as a failure."
 - ▶ Placebo: shifting the target circularly (250–3,000 days) breaks its alignment with the features but keeps each series' own dependence
 - ▶ a valid null for $\max_j |t_j|$ if the target is stationary and the shifts exceed its dependence range
- **Critique**: "Act as a referee: which of these significant features are one effect counted several times, and which are known anomalies rather than new ones?"
- **Report**: all M tests, all trials including the discarded ones, DSR and PBO for any strategy built on them

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